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Thesis submitted in fulfilment of the requirements
for the award of the degree of
Doctor of Philosophy of Engineering in Chemical

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JANUARY 2013

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Acknowledgement is a statement of gratitude for assistance in producing a work. Usually the first person you have to acknowledge is your main supervisor or Professor. Follow by your co-supervisor. You may also include those who have help or assist you in completing your research e.g. other Professors, lab technician, lab members etc.

Other than that, authors also often thank their family, wife, children or maybe friends in their acknowledgement. Anyhow acknowledgement is a part when you would like to express to sincere gratitude to those who helped you in completing your work.

Assoc. Prof. Ir. Dr.-Ing. Mohamad Rizza bin Othman *PEng CEng MIChemE*
UMP Gambang
21 April 2020

ABSTRAK

Abstrak adalah ringkasan pendek penyelidikan siap. Jika dilakukan dengan baik, ia membuat pembaca ingin mengetahui lebih lanjut mengenai penyelidikan anda. Komponen asas abstrak dalam sebarang disiplin termasuk motivasi / pernyataan masalah, kaedah / prosedur / pendekatan, keputusan / penemuan / produk dan kesimpulan / implikasi. Motivasi / kenyataan masalah: Mengapa kita mengambil berat tentang masalah ini? Apakah jurang praktikal, saintifik, teori atau seni adalah pengisian penyelidikan anda? Kaedah / prosedur / pendekatan: Apa yang kamu lakukan untuk mendapatkan keputusan anda? (contohnya dianalisis 3 novel, menyiapkan satu siri 5 lukisan minyak, ditemu bual 17 pelajar) Keputusan / penemuan / produk: Sebagai hasil melengkapkan prosedur di atas, apakah yang dilakukan oleh anda belajar / mencipta / buat? Kesimpulan / implikasi: Apakah implikasi yang lebih besar penemuan anda, terutamanya bagi masalah/jurang yang dikenal pasti dalam langkah 1? Walau bagaimanapun, ia adalah penting untuk mengambil perhatian berat yang diberikan kepada komponen yang berbeza boleh berbeza-beza mengikut disiplin. Bagi model, cuba untuk mencari abstrak penyelidikan yang serupa dengan penyelidikan anda.

ABSTRACT

An abstract is a short summary of your completed research. If done well, it makes the reader want to learn more about your research. The basic components of an abstract in any discipline should include motivation / problem statement, methods / procedure / approach, results / findings / product and conclusion / implications. Motivation/problem statement: Why do we care about the problem? What practical, scientific, theoretical or artistic gap is your research filling? Methods / procedure / approach: What did you actually do to get your results? (e.g. analyzed 3 novels, completed a series of 5 oil paintings, interviewed 17 students) Results/findings/product: As a result of completing the above procedure, what did you learn/invent/create? Conclusion/implications: What are the larger implications of your findings, especially for the problem/gap identified in step 1? However, it's important to note that the weight accorded to the different components can vary by discipline. For models, try to find abstracts of research that is similar to your research.

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LIST OF SYMBOLS

λ	largest eigenvalue
a	alternative
i	criteria
I	score index
j	indicator
k	number of alternatives
n	priority value of subordinates
p	priority value of the parent's level
PV	priority value
V	assessment score
v	assessment value
V_N	normalized assessment score
w	eigenvector
A	matrix of pairwise comparison
W	supermatrix
T_a	highest assessment value margin
V_{pos}	positive assessment score

List of Sub/Superscripts and Operations

a	alternative
c	chemical component
cp	chemical mass process
ep	energy generate process
FCC	fixed capital cost
out	output
p	product

LIST OF ABBREVIATIONS

AHP	Analytic hierarchy process
ANP	Analytic network process
CEI	Chemical exposure index
CEO	Chief executive officer
DfS	Design for sustainability
ESH	Environment, health and safety
HAZOP	Hazard and operability study
LCA	Life cycle analysis
PDM	Process decision making
PSE	Process system engineering

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CHAPTER 1

INTRODUCTION

1.1 Research Background

The rapid paradigm shift toward high-level autonomous systems is fundamentally transforming global industrial modalities, establishing new benchmarks for operational efficiency, safety, and productivity across diverse sectors (Guizzo, 2008). Within this broader context, off-road autonomous navigation represents an especially critical and challenging deployment domain, particularly for heavy agricultural machinery tasked with operating autonomously within unstructured and severe field environments (Hameed, Bochtis, et al., 2021). Unlike highly regulated urban streets or uniform industrial indoor settings, unstructured natural terrains present multi-faceted complexities, including irregular surface roughness, unpredictable slope variations, highly irregular topographies, and dynamic obstacles (Sofman, Lin, et al., 2006). To guarantee reliable vehicle state estimation and successful terrain traversal under these erratic conditions, the integration of robust, terrain-aware path planning architectures represents an indispensable algorithmic prerequisite (Sedighi, Nguyen, et al., 2019). Consequently, overcoming these geometric terrain hazards stands as a primary engineering challenge in modern mobile robotics design.

Within the specific domain of precision agriculture, autonomous navigation frameworks must directly resolve acute demographic, economic, and operational constraints. The global agricultural sector is currently facing a dual challenge: an escalating demand for food and crop products driven by population growth, occurring alongside a severe, long-term contraction in the available skilled labor pool (Emmi & Gonzalez-de Santos, 2014). To mitigate these macro-economic vulnerabilities, increasing the scale of dependable in-field automation and mobile robotics platforms has become a structural necessity (Emmi & Gonzalez-de Santos, 2014). Effective path planning algorithms designed for these agricultural platforms must look past simple binary (passable/impassable) obstacle avoidance; they must concurrently minimize mechanical energy dissipation, optimize total travel time, and enforce absolute platform stability over highly undulating landscapes (Hameed, Bochtis, et al., 2021). To systematically address these multi-objective criteria, this study focuses on the development, implementation, and validation of a comprehensive 2.5D terrain-aware path planning framework explicitly engineered to optimize autonomous ground vehicle routing across challenging off-road and agricultural topogra-

phies.

1.2 Background of the Study

Path planning represents a foundational architectural pillar of mobile robot autonomy, enabling automated platforms to execute localized routing maneuvers deterministically. Mechanistically, this process is tasked with computing an optimal or near-optimal trajectory from an initial coordinates origin to a targeted terminal destination while concurrently satisfying explicit kinematic constraints and avoiding localized environmental obstacles (Sánchez-Ibáñez, Pérez-del Pulgar, et al., 2021). Historically, classical graph-search and grid-allocation strategies, most notably the Dijkstra and A* algorithms, were pioneered to resolve these pathfinding problems within planar, two-dimensional (2D) static environments (Karur, Sharma, et al., 2021; Noto & Sato, 2000). These foundational frameworks operate by evaluating discrete traversal costs between adjacent nodes distributed across a structured grid network or topological graph representation, typically optimizing the search space to minimize cumulative path lengths or similar spatial clearance metrics (Karur, Sharma, et al., 2021; Noto & Sato, 2000). For example, the deterministic search behavior of the standard A* algorithm is governed by a multi-variant evaluation cost function formalized as:

$$f(n) = g(n) + h(n) \quad (1.1)$$

where $g(n)$ defines the exact accumulated cost computed from the initial start node to the current node n , and $h(n)$ represents the heuristic estimation of the remaining traversal cost required to reach the terminal goal state (Karur, Sharma, et al., 2021; Noto & Sato, 2000). While these planar graph-search heuristics provide global mathematical optimality within flat workspace boundaries, their operational efficacy degrades significantly when deployed across non-planar topographies.

Traditional A* path planning paradigms fall short when confronted with the complex, uneven terrains typical of off-road and agricultural settings because they inherently disregard the vertical elevation dimension (Bettencourt, Caramalho, et al., 2025). This geometric omission can lead to severe navigation failures, such as catastrophic collisions with unmodeled structural obstacles positioned physically above the robot's primary 2D reference plane, or the generation of global paths that are physically untraversable due to excessive, unsafe slope gradients (Bettencourt, Caramalho, et al., 2025). To resolve these spatial deficiencies without incurring prohibitive processing overheads, two-and-a-half-dimensional (2.5D) environmental representations have emerged as a highly practical structural compromise between full 3D volumetric voxel complexity and 2D planar simplicity. Mathematically, a 2.5D elevation map correlates horizontal planar coordi-

nates (x, y) with a single, discrete altitude value z for every coordinate cell, successfully capturing localized terrain height variations and allowing the system to derive key topographic features like slope gradients (Bettencourt, Caramalho, et al., 2025). This strategic abstraction enables pathfinding algorithms to directly factor in critical traversability criteria and localized step distances, completely bypassing the extreme computational burdens and data densities associated with full 3D volumetric representations (Bettencourt, Caramalho, et al., 2025). Transitioning to these 2.5D mapping models is particularly relevant for autonomous agricultural vehicles, where executing stable navigation across sloped landscapes and unmanaged crop fields requires a highly nuanced, topologically aware understanding of terrain profiles to prevent platform instability (Hameed, Bochtis, et al., 2021).

1.3 Problem Statement

The primary challenge in off-road and agricultural navigation stems from the dynamic, non-linear complexities of uneven terrain, which cannot be adequately modeled or predicted by conventional two-dimensional (2D) path planning paradigms (Bettencourt, Caramalho, et al., 2025). Relying exclusively on 2D planar configurations risks generating trajectories that are locally optimal in terms of horizontal distance but physically impossible, highly inefficient, or dangerous due to severe slope gradients, large elevation changes, or undetected vertical obstacles (Bettencourt, Caramalho, et al., 2025). For instance, a planar navigation routine may execute a command leading to a terminal collision if it fails to evaluate obstacles positioned above its active sensor observation plane, inducing severe tracking and navigation errors (Bettencourt, Caramalho, et al., 2025). Furthermore, attempting to retroactively fix these geometric oversights via standard path post-processing often fails to resolve fundamental physical incompatibilities, yielding discontinuous, “choppy” trajectories or acute turning angles that present severe vehicle control challenges (Frey, Hoeller, et al., 2022). Consequently, relying on flat surface assumptions directly jeopardizes vehicle safety and structural integrity on natural topographies.

Existing off-road path planning methodologies frequently operate within a restrictive trade-off, either oversimplifying the underlying terrain model to preserve computational velocity or employing full three-dimensional (3D) configurations that incur prohibitive computational overhead (Overbye, 2022). While widespread modifications to the classical A* graph-search algorithm successfully optimize runtime search velocities, minimize raw path lengths, or manage transient dynamic obstacles (Injarapu & Gawande, 2023; Lei, Zhang, et al., 2021; Li & Hu, 2020), they fail to systematically integrate multi-dimensional terrain characteristics into the core cost-evaluation loop for proactive decision-making. Specifically, within the domain of agricultural robotics, the direct, real-time eval-

uation of physical three-dimensional step distances (3DSD) and localized slope magnitudes (SM) within the internal A* search mechanism, compounded by an adjustable slope weight, remains a critical research gap requiring rigorous investigation. Additionally, standard trajectory smoothing techniques routinely execute geometric alterations without enforcing compliance with hard environmental traversability boundaries, such as the vehicle’s Maximum Climbable Slope (MCS). This algorithmic omission results in smoothed paths that look visually appealing but violate the platform’s actual physical climbing tolerances.

Therefore, the core engineering problem addressed by this research is the absence of a unified path planning framework that effectively leverages the computational balance and detail offered by a 2.5D terrain representation. Resolving this problem necessitates the direct, intelligent embedding of key 2.5D structural terrain properties directly into the node-selection logic of a modified A* algorithm, ensuring the proactive generation of intrinsically traversable routes. This optimization must be seamlessly integrated with a kinematically validated trajectory smoothing architecture that rigorously respects vehicle operating limits. Successfully executing this integrated strategy is vital to enabling safe, energy-efficient, and truly autonomous navigation across challenging agricultural environments.

1.4 Research Gap

Despite extensive advancements in autonomous vehicle navigation and graph-search adaptations, several critical gaps persist regarding the integrated frameworks required for safe, efficient, and terrain-aware navigation in complex off-road and agricultural settings. This overarching deficiency in contemporary field robotics literature is characterized by three distinct technical gaps:

Firstly, a significant gap remains concerning the systematic integration of detailed three-dimensional geometric terrain attributes directly within the core cost-evaluation loops of two-and-a-half-dimensional (2.5D) path planning algorithms. While the classical A* algorithm has seen numerous architectural enhancements designed to optimize runtime search efficiency, manage binary obstacle avoidance, and navigate transient dynamic environments (Injarapu & Gawande, 2023; Lei, Zhang, et al., 2021; Li & Hu, 2020), fewer studies explicitly embed a comprehensive set of 2.5D terrain features, such as varying slope magnitudes (SM) and precise 3D step distances (3DSD), directly into its cost function for agricultural applications. Existing 2.5D navigation methods predominantly focus on discrete obstacle representation and structural layout mapping rather than nuanced traversability costs derived directly from continuous terrain geometry. Furthermore, contemporary frameworks frequently utilize generic or static cost indices that fail to dy-

namically adapt to the highly variable topographies and specific operational challenges encountered by agricultural robots (Bettencourt, Caramalho, et al., 2025). Consequently, the development of a unified multi-terrain cost function that proactively guides the graph search based on these physical terrain properties and an adjustable slope weight (λ) represents an unaddressed and highly necessary contribution.

Secondly, a secondary research gap involves the lack of boundary constraint validation and climbability threshold enforcement within post-processing path smoothing techniques. Standard path optimization protocols widely implement geometric refinement methods, such as B-splines and Bezier curves, to elevate path aesthetics, eliminate acute turning variations, and improve vehicle drivability (Frey, Hoeller, et al., 2022; Han, Xu, et al., 2019; Zhang & Li, 2021). However, a significant deficiency exists in consistently coupling these continuous geometric smoothing processes with a rigorous validation of terrain- climbability thresholds (e.g., Maximum Climable Slope) and other localized boundary constraints. Because typical smoothing algorithms focus exclusively on minimizing curvature derivatives, they risk generating kinematically fluid trajectories that, while visually appealing, inadvertently route the vehicle through steep sections that violate the actual physical traversability limits of the platform, leading to unsafe or impossible execution profiles (Frey, Hoeller, et al., 2022). Currently, the explicit coupling of B-spline refinement with discrete sampling checks for boundary compliance, vegetation exposure, and maximum slope thresholds is not extensively addressed in the literature for 2.5D terrain-aware planning in agricultural environments. Proactively embedding these physical threshold checks immediately following or during the smoothing loop is therefore essential to prevent vehicle instability.

Finally, existing literature exhibits a pronounced empirical gap characterized by the absence of a comprehensive comparative analysis contrasting terrain-aware 2.5D graph-search algorithms directly against conventional 2D baselines across highly diverse topographic profiles. While performance evaluation for path planning algorithms is a standard convention in mobile robotics research, contemporary frameworks frequently rely on oversimplified geometric metrics or fail to provide a direct, side-by-side comparison with a traditional 2D baseline to demonstrate the explicit value of 2.5D-informed cost functions (Hameed, Bochtis, et al., 2021; Song, Liu, et al., 2019). This empirical deficiency is particularly apparent across heterogeneous terrain environments, where algorithms must be benchmarked under highly variable gradient complexities, including low-gradient plains, medium-relief hills, and high- relief escarpments. Furthermore, a comprehensive suite of performance parameters—comprising exact 3D path lengths, travel-time proxies, computational execution runtimes, and mechanical energy consumption proxies—is rarely evaluated simultaneously. Addressing this empirical shortfall via a rigorous comparative evaluation, conducted across a spectrum of adjustable slope weight (λ) settings, is necessary to robustly quantify the operational benefits and edge-computing viability of the proposed

approach.

1.5 Research Objectives

The primary objective of this research is to design, implement, and validate a comprehensive, safety-conscious navigation architecture tailored for autonomous ground vehicles (AGVs) operating across unstructured environments. To achieve this macro-level goal, the study is systematically segmented into three primary technical objectives:

1. To design and formulate a novel, two-and-a-half-dimensional (2.5D) A* graph-search path planning algorithm that directly integrates physical three-dimensional step distances (3DSD), localized slope magnitudes (SM), and an adjustable slope weight parameter (λ) into a unified multi-terrain cost function. This development aims to move beyond planar simplifications by directly embedding multi-dimensional geometric terrain hazards into the core node-selection heuristic, enabling the pathfinder to proactively avoid high-risk sloped regions during the global graph search phase.
2. To implement a post-processing B-spline refinement module engineered to smooth sharp, discrete turning variations characteristic of grid-based node expansion while strictly validating terrain-climbability thresholds and localized boundary constraints. This architectural module ensures that continuous geometric optimization routines are strictly bound by the platform's physical operating thresholds, such as its Maximum Climable Slope (MCS), thereby preventing the generation of kinematically fluid but physically untraversable routes.
3. To verify, evaluate, and benchmark the integrated path planning framework across diverse synthetic terrain profiles against a conventional planar two-dimensional (2D) baseline planning architecture. This empirical verification protocol is designed to simultaneously record and analyze a comprehensive suite of performance indicators—specifically total path length parameters, travel-time proxies (TTP), algorithmic execution runtimes, and mechanical energy consumption metrics—to robustly quantify the operational efficiency and edge-computing viability of the proposed framework.

1.6 Research Questions

To systematically achieve the stated research objectives and address the identified gaps in field robotics literature, this study is guided by three core research questions that isolate the algorithmic, kinematic, and empirical dimensions of the navigation framework:

1. **Algorithmic Integration:** How can a two-and-a-half-dimensional (2.5D) A* path planning algorithm effectively integrate physical three-dimensional step distances (3DSD), localized slope magnitudes (SM), and an adjustable slope weight parameter (λ) into a unified multi-terrain cost function to ensure proactive obstacle and hazard avoidance during off-road navigation?
2. **Kinematic Refinement & Validation:** What is a mathematically robust and computationally efficient method for implementing a post-processing B-spline refinement module to smooth sharp, discrete path turns while strictly validating terrain-climbability thresholds and environmental boundary constraints across a 2.5D terrain-aware trajectory?
3. **Empirical Benchmarking:** How does the proposed terrain-aware framework perform when verified and evaluated across diverse synthetic terrain profiles against a conventional planar A* baseline in terms of total three-dimensional path length (L_{3D}), speed-adjusted travel-time proxies (TTP), algorithmic execution runtime, and uphill-selective energy consumption proxies (ECP)?

1.7 Scope of the Study

The operational and architectural boundaries are systematically constrained across specific spatial mapping frameworks and algorithmic parameters to ensure a focused validation of the navigation framework. Mechanistically, the environmental mapping pipeline is limited to the conversion of simulation-derived three-dimensional (3D) point cloud data, stored in .pcd file formats, into static two-and-a-half-dimensional (2.5D) elevation grids featuring a single-layer vegetation penalty mask. Upon this discrete grid infrastructure, the core pathfinding system implements an 8-connected grid A* algorithm designed to execute global graph-search operations. During the runtime node-expansion phase, this modified pathfinder directly evaluates physical three-dimensional step distances (3DSD), localized slope magnitudes (SM), and a manually adjusted slope weight parameter (λ) within its internal cost-calculation loop. By restricting the mapping inputs and node-evaluation metrics to these explicit grid parameters, the framework establishes a highly structured digital workspace for deterministic global path generation.

Beyond the initial graph-search parameters, the post-processing trajectory routines, empirical evaluation thresholds, and validation methodologies are strictly bounded to isolate safety and efficiency performance metrics. Trajectory optimization is restricted to a post-processing path refinement module utilizing open-uniform B-splines, engineered to smooth sharp, discrete turning variations characteristic of 8-connected grid node transitions. To guarantee absolute physical traversability, this continuous geometric smoothing

is coupled with discrete sampling checks that rigorously validate environmental boundary compliance, vegetation exposure constraints, and maximum slope thresholds. The empirical evaluation limits are defined by testing the navigation framework across three distinct synthetic terrain profiles—specifically low-gradient, medium-relief, and high-relief topographies—using five specific slope weight configurations and three unique start- goal coordinate pairs per map environment. In alignment with safe development and prototyping protocols, all experimental trials, terrain modeling operations, and trajectory visualizations are conducted exclusively through simulation within a controlled virtual environment, without physical vehicle deployment. This comprehensive bounding of evaluation metrics and virtual simulation boundaries ensures a highly controlled, repeatable testing sandbox necessary to robustly verify the computational performance of the modified path planning system.

1.8 Significance of the Study

The theoretical insights, algorithmic architectures, and practical applications generated by this research hold significant implications for the advancement and safe deployment of autonomous ground vehicles operating across unstructured environments. By resolving the critical geometric and safety limitations inherent in traditional planar pathfinding, this study bridges the operational gap between high-level global navigation and low-level physical platform stability on non-planar topographies. The distinct contributions of this work are systematically structured across theoretical, practical, and methodological dimensions to offer a scalable foundation for modern field robotics.

The theoretical significance of this research lies in its targeted advancement of the state-of-the-art in multi-dimensional off-road routing paradigms. This study introduces a novel, unified multi- terrain cost function designed specifically for a modified 2.5D A* graph-search algorithm. By directly embedding physical three-dimensional step distances and localized slope magnitudes into the core cost-evaluation loop, the framework provides a more realistic, physically informed pathfinding approach than conventional planar methods. Furthermore, the development of a post-processing B-spline smoothing module that incorporates stringent boundary validation against terrain- climbability thresholds contributes to the generation of trajectories that are not only kinematically smooth but also inherently safe and physically traversable. This integrated mathematical approach directly resolves the traditional limitations of unconstrained geometric smoothing and flat surface assumptions, establishing a robust, reproducible methodology for navigation across complex terrain profiles.

Beyond these theoretical contributions, the practical applications of this framework directly enhance the operational safety, efficiency, and sustainability of autonomous

machinery deployed in precision agriculture. By implementing a terrain-adaptive path planning loop, the framework mitigates severe operational risks, leading to enhanced safety metrics by reducing vehicle exposure to catastrophic rollovers, collisions, or immobilization on untraversable gradients (Li, Tang, et al., 2019). Factoring the Maximum Climable Slope (MCS) directly into the optimization pipeline as a hard traversability constraint acts as a critical safety barrier to protect expensive, capital-intensive field machinery and prevent localized crop damage (Saeed, Omri, et al., 2022). Concurrently, optimizing paths based on 3D distances, slopes, and speed-adjusted travel-time proxies (TTP) translates into minimized cycle times and elevated productivity during field execution (Hameed, Bochtis, et al., 2021). Operational sustainability is further reinforced by proactively evaluating uphill-selective energy consumption proxies (ECP) during the graph search, which directly minimizes cumulative fuel consumption and battery depletion rates (Hameed, Bochtis, et al., 2021). Additionally, the system’s capacity to ingest static 2.5D elevation grids derived from point cloud data and overlay a single-layer vegetation penalty mask ensures a high degree of adaptability across diverse agricultural landscapes, ranging from standard crop fields to heavily sloped terrains.

Finally, this research delivers critical methodological advancements by establishing a highly systematic and cost-effective validation pipeline for autonomous vehicle benchmarking. The simulation-only approach allows for extensive, repeatable experimentation across a broad spectrum of heterogeneous synthetic terrain profiles—specifically low-gradient plains, medium-relief hills, and high-relief escarpments—completely bypassing the severe logistical constraints, equipment costs, and physical hazards of real-world prototyping trials. Utilizing a comprehensive performance suite comprising exact 3D path lengths (L_{3D}), travel-time proxies (TTP), computational execution runtimes, and energy consumption proxies (ECP) provides a rigorous mathematical baseline for direct comparison against traditional 2D planners. Benchmarking these metrics across a variable range of slope weight configurations offers transparent, quantifiable insights into the precise efficiency and safety gains achieved by the terrain-aware approach. This methodological rigor ensures that the resulting data trends remain robustly scalable and mathematically sound for future physical implementations.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The accelerating demand for high-level autonomous capabilities within unstructured environments, particularly across off-road and precision agricultural settings, has catalyzed extensive algorithmic developments within advanced path planning and navigation architectures (Hameed, Bochtis, et al., 2021). As established in Chapter 1, navigating complex topographies requires an explicit transition away from planar two-dimensional (2D) path planning paradigms toward intelligent, terrain-aware routing frameworks. To construct a rigorous theoretical foundation for this algorithmic shift, a systematic evaluation of existing literature is necessary to contextualize how field robotics platforms perceive, evaluate, and traverse rough terrain. Consequently, this introductory section sets the analytical scope for a comprehensive review of the historical milestones and contemporary methodologies that govern off-road mobile robot autonomy.

The systematic review of literature within this chapter is structurally organized across several interconnected operational and algorithmic sub-systems to frame the subsequent research methodology. It explores foundational and contemporary work regarding the physical challenges of off-road navigation, the mathematical design of two-and-a-half-dimensional (2.5D) terrain representations, and the architectural evolution of the A* graph-search family. Furthermore, this chapter dissects established techniques for post-processing trajectory smoothing and localized constraint validation alongside the empirical benchmarking metrics utilized for performance evaluation. By critically analyzing these current methodologies to isolate areas of consensus, divergence, and operational limitations, this review serves to clearly delineate the specific technical gaps that this Master's thesis addresses. Grounding the study within this established scholarly landscape provides a clear justification for the development of the proposed terrain-aware path planning framework.

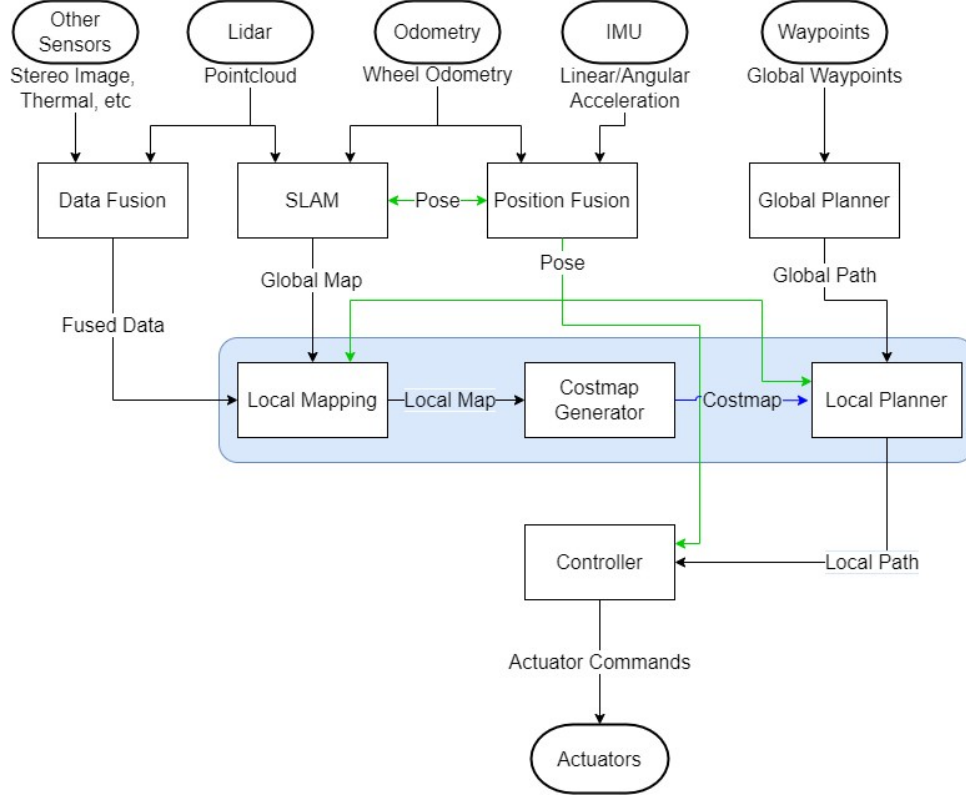


Figure 2.1: System overview of a typical unmanned ground vehicle.

2.2 Off-Road and Agricultural Autonomous Navigation

2.2.1 Challenges in Off-Road Environments

Off-road environments inherently present a significantly more complex and dynamic challenge for autonomous navigation systems compared to highly regulated, structured urban settings (Sofman, Lin, et al., 2006). These unstructured spaces are characterized by unpredictable macro-topographical variations, including severe slope gradients, uneven surface profiles, and a dense distribution of heterogeneous environmental obstacles (Sofman, Lin, et al., 2006). Specifically, an autonomous vehicle must continuously perceive and negotiate a wide array of static and dynamic anomalies—such as large rocks, structural depressions, and dense field vegetation—that disrupt steady traction and clear path traversal (Sofman, Lin, et al., 2006). Consequently, managing these irregular environmental profiles requires a transition toward advanced navigation architectures capable of dynamically interpreting multi-dimensional workspace geometries.

Traditional 2D path planning architectures are fundamentally unsuited for these rough topographies because they operate on a simplified planar representation of the environment that completely neglects the crucial vertical dimension (Bettencourt, Caramalho, et al., 2025). This structural oversight routinely leads to the generation of global tra-

jectories that are geometrically optimal in terms of horizontal distance but physically untraversable, unsafe, or highly inefficient when projected onto uneven ground (Bettencourt, Caramalho, et al., 2025). For example, a planar navigation routine lacks the spatial mapping capacity to detect overhead clearance obstacles positioned above its observation plane or may guide a vehicle across a steep incline that exceeds the platform’s physical climbing capabilities, resulting in catastrophic structural collisions or complete vehicle immobilization (Bettencourt, Caramalho, et al., 2025). Therefore, executing safe and efficient off-road navigation necessitates a robust mathematical understanding and direct integration of three-dimensional terrain characteristics into the core path planning process.

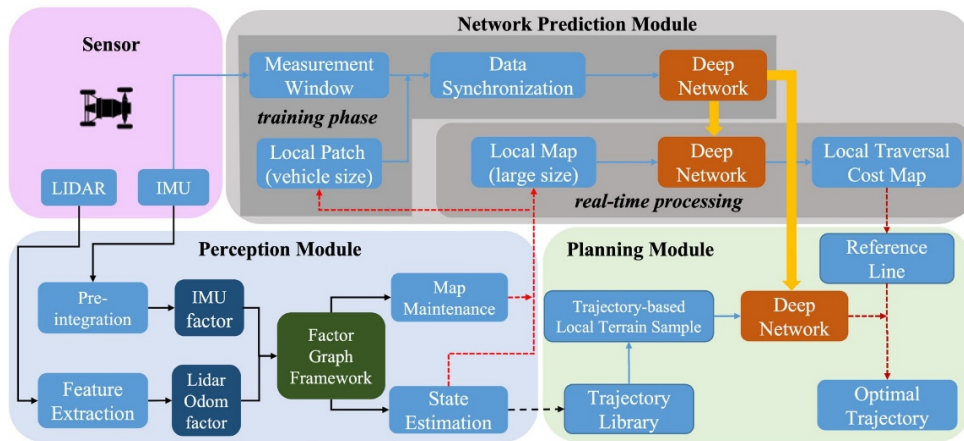


Figure 2.2: Architecture of the proposed autonomous navigation approach. During real-time navigation, the perception module provides the original point cloud map and the vehicle pose. The prediction module predicts the local point cloud map and performs traversal cost.

2.2.2 Autonomous Systems in Agriculture

The ongoing integration of autonomous navigation technologies within the agricultural sector is primarily driven by expanding global food demands and an acute, long-term contraction of the skilled rural labor force (Emmi & Gonzalez-de Santos, 2014). To address these production bottlenecks, autonomous agricultural ground robots are increasingly being deployed across field operations to execute complex tasks such as precision seeding, automated harvesting, and broad-acre field surveillance (Algfoor, Sunar, et al., 2013; Hameed, Bochtis, et al., 2021). However, these unique field deployment scenarios impose rigorous and specialized architectural requirements on motion planning frameworks that extend far beyond standard urban navigation paradigms. Beyond simply achieving collision avoidance, agricultural mobile platforms must traverse complex crop layouts and undulating sloped landscapes both resource-efficiently and safely (Hameed, Bochtis, et al., 2021). This multidimensional operational profile necessitates optimiza-

tion across a diverse suite of competing performance criteria, including minimizing travel time constraints, reducing cumulative mechanical energy expenditure, and enforcing rigid platform stability on varying terrains to actively prevent catastrophic rollover accidents (Hameed, Bochtis, et al., 2021; Li, Tang, et al., 2019). Consequently, achieving high operational precision during in-field maneuvers underscores the vital importance of deploying terrain-aware path planning systems that explicitly account for localized topographic variations and variable surface conditions.

2.2.3 Positioning Systems for Off-Road Vehicles

Reliable three-dimensional (3D) navigation and deterministic path execution for autonomous off-road vehicles rely fundamentally on high-precision localization architectures. Modern state estimation suites typically fuse complementary sensor modalities to achieve robust, real-time vehicle positioning across unstructured terrains. For instance, Differential Global Positioning Systems (DGPS) provide absolute global coordinates, whereas Inertial Navigation Systems (INS) deliver high-frequency attitude rates and localized motion data (Xiang & Ozguner, 2006). The algorithmic integration of these asynchronous DGPS and INS measurements is commonly executed utilizing state-space architectures such as Extended Kalman Filters (EKF), which can be structurally deployed via either an error state space formulation or a total state space formulation (Xiang & Ozguner, 2006). While the total state space approach provides an optimally weighted blending of multi-sensor data profiles, its implementation introduces high computational iteration rates and increases systemic vulnerability to individual sensor failures (Xiang & Ozguner, 2006). These multi-sensor fusion frameworks significantly attenuate sensor noise and elevate the estimation accuracy of the vehicle’s position, pitch, roll, and height configurations, which are critical for constructing high-fidelity environmental terrain maps and tracking planned global trajectories (Hameed, Bochtis, et al., 2021; Xiang & Ozguner, 2006).

2.3 Terrain Representation and Mapping for Off-Road Navigation

2.3.1 2D vs. 3D vs. 2.5D Terrain Models

The selection of an optimal environmental mapping paradigm represents a critical design choice in off-road path planning, directly influencing an algorithm’s computational complexity, geometric accuracy, and physical applicability. Traditional 2D grid maps simplify the workspace configuration into a flat plane, rendering global path generation highly

efficient from a computational standpoint. Under this planar assumption, classical graph-search algorithms operate effectively within low-overhead tracking boundaries (Karur, Sharma, et al., 2021). However, this flat abstraction is fundamentally limited when deployed across uneven or natural landscapes because it completely ignores crucial vertical geometry, routinely leading to the generation of physically untraversable paths, structural collisions, and catastrophic navigation errors (Bettencourt, Caramalho, et al., 2025). Consequently, relying on flat surface assumptions strips the navigation system of its ability to proactively identify macro-topographical variations before path execution.

To overcome the spatial blindness of planar models, full 3D volumetric representations can be utilized to capture the complete, high-fidelity geometry of an unstructured environment. These multi-dimensional mapping frameworks preserve the complete spatial topology of the workspace, enabling comprehensive obstacle detection, negative obstacle mapping, and detailed traversability analysis. While ideal in theoretical formulations, full 3D representations introduce massive data densities and high geometric dimensionality that incur severe computational overhead (Overbye, 2022). This processing burden demands substantial runtime memory storage and processing power, making real-time path planning exceptionally challenging for computationally restricted embedded autonomous platforms (Overbye, 2022). As a result, the extreme processing latencies associated with dense volumetric data structures often limit their practical application in dynamic edge-computing scenarios.

Two-and-a-half-dimensional (2.5D) terrain models have emerged as a highly practical structural compromise, balancing the computational simplicity of 2D layouts with the rich geometric fidelity of full 3D configurations. This mapping paradigm correlates horizontal planar coordinates (x, y) with a single, discrete altitude value z for each individual grid cell, successfully preserving critical height variations and continuous surface characteristics (Gu, Cao, et al., 2011; Xie, Fang, et al., 2020). This coordinate abstraction allows pathfinding algorithms to execute direct, real-time evaluations of localized slope magnitudes (SM) and physical three-dimensional step distances (3DSD), which are critical for accurately quantifying traversability and calculating instantaneous movement costs (Bettencourt, Caramalho, et al., 2025; Xie, Fang, et al., 2020). By integrating these vital vertical dimensions without inheriting full volumetric voxel densities, 2.5D configurations provide sufficient geometric detail for safe, resource-efficient off-road routing. This structural balance is particularly vital for autonomous agricultural vehicles, where executing stable navigation across sloped landscapes and unmanaged crop fields requires a highly nuanced, topologically aware understanding of terrain profiles to prevent platform instability (Hameed, Bochtis, et al., 2021).

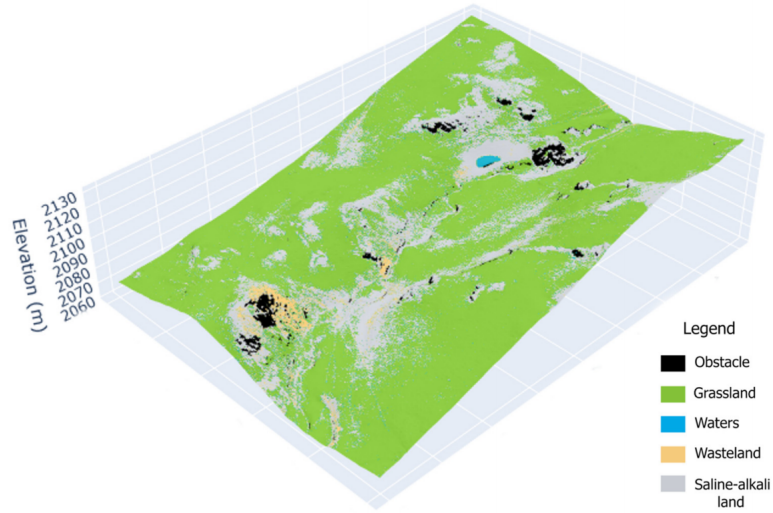


Figure 6. Real-world 3D environment modeling results.

Figure 2.3: Real-world 3D environment modeling results.

2.3.2 Point Cloud Data and Grid-Based Mapping

The systematic acquisition and processing of spatial environmental data represent the foundational phase in transforming raw, unstructured sensory inputs into actionable geometric matrices for mobile robot localization and path generation. This mapping pipeline predominantly begins with the collection of dense three-dimensional (3D) point cloud datasets, typically stored in .pcd file formats and captured via high-resolution Light Detection and Ranging (LiDAR) sensor suites (Himmelsbach, Müntzinger, et al., 2021). To be utilized by global graph-search configurations, these unorganized spatial coordinate vectors must be mathematically parsed, filtered, and transformed into structured, low-overhead topologies, which for 2.5D terrain representations entails converting the volumetric points into a static elevation grid map (Xie, Fang, et al., 2020). This coordinate transformation is typically executed by discretizing a horizontal reference plane—such as the XY plane—into a network of uniform square cells, and subsequently approximating the localized 3D environmental geometries and structural obstacles using discrete bounding cuboids parameterized by the minimum and maximum vertical coordinate values (e.g., the Z -axis) found within each grid boundary (Xie, Fang, et al., 2020,?). Furthermore, spatially overlapping obstacles occupying a single cell coordinate are algebraically merged into a unified layer, establishing a definitive traversable domain bounded directly by the upper surface facets of these approximated bounding boxes (Xie, Fang, et al., 2020). This geometric projection technique compresses high-density volumetric point clouds into a streamlined, non-planar heightfield grid optimized for real-time pathfinding operations.

Beyond the mapping of rigid geometric macro-topographies, executing safe navigation within dense agricultural environments requires superimposing non-geometric se-

mantic features onto the underlying spatial layout. To achieve this layered environmental awareness, the integration of a specialized single-layer vegetation penalty mask is necessary to model sections of the workspace that, while physically passable, introduce substantial locomotion resistance or risk inducing structural crop damage (Himmelsbach, Müntzinger, et al., 2021). These non-binary cost layers operate by dynamically injecting scalar penalty values into coordinate cells that exhibit significant vegetation height or dense canopy distributions, thereby artificially inflating the local traversal cost during node expansion. By manipulating the heuristic search in this manner, the global path planner is proactively guided away from high-resistance, vegetative zones and steered toward cleaner, unencumbered paths. Layering these semantic cost metrics onto the baseline geometric elevation grid ensures that the generated trajectories respect both the hard kinematic limits of the terrain and the soft operational safety constraints of the agricultural environment.

2.4 Path Planning Algorithms for Mobile Robots

2.4.1 Heuristic Search Algorithms (A* and its Variants)

Heuristic search algorithms, particularly the A* algorithm, form the foundational bedrock of mobile robot path planning systems due to their proven mathematical optimality and operational efficiency within static environments (Karur, Sharma, et al., 2021; Noto & Sato, 2000). Mechanistically, the algorithm systematically explores a graph or a discretized grid network to compute a trajectory connecting an initial start node to a terminal goal state with the absolute minimum accumulative cost. To govern this deterministic graph search, A* implements a multi-variant evaluation function expressed as:

$$f(n) = g(n) + h(n) \quad (2.1)$$

where $g(n)$ represents the exact, accumulated cost traversed from the initial start node to the current node n (Karur, Sharma, et al., 2021; Noto & Sato, 2000; Zhang & Li, 2021). Concurrently, $h(n)$ acts as a heuristic function estimating the remaining traversal cost from node n to the terminal goal state (Karur, Sharma, et al., 2021). For the algorithm to preserve its mathematical guarantees of optimality and completeness, the design of $h(n)$ must remain strictly admissible—meaning it never overestimates the true remaining cost—and mathematically consistent (Yao, Lin, et al., 2010). This core cost infrastructure underpins the node selection strategy executed during live spatial exploration.

At an execution level, the graph search manages node categorization using two synchronized data structures known as the Open List and the Closed List (Baskara, Ri-

vai, et al., 2019; Chen, Kong, et al., 2013; Duc Dung, Thai Duc, et al., 2021; Zhao, Li, et al., 2016). The algorithm operates through iterative optimization loops, continuously selecting the specific node that exhibits the lowest $f(n)$ value from the Open List, transitioning it to the Closed List, and evaluating its adjacent neighboring cells (Baskara, Rivai, et al., 2019; Chen, Kong, et al., 2013). To configure the structural topology of the grid network, implementing an 8-connected grid mapping paradigm permits both orthogonal and diagonal node transitions. This multi-directional connectivity offers superior spatial flexibility and generates significantly shorter path metrics than restrictive 4-connected grid allocations. The geometric behavior of these spatial transitions is directly tied to the mathematical distance equations selected for the heuristic estimator.

The localized search velocity and directional bias of the pathfinder are heavily governed by the specific spatial distance formula chosen to represent $h(n)$. For grid configurations where unconstrained multi-directional traversal is permitted, the Euclidean distance represents the absolute straight-line metric calculated as:

$$h_{Euclidean}(n) = \sqrt{(x_{goal} - x_n)^2 + (y_{goal} - y_n)^2} \quad (2.2)$$

which serves as a standard baseline across diverse pathfinding systems (Karur, Sharma, et al., 2021; Zhang & Li, 2021; Zhuang, Zhang, et al., 2019). Alternatively, when vehicle translations are constrained to standard grid lines, the Manhattan distance maps orthogonal displacements via:

$$h_{Manhattan}(n) = |x_{goal} - x_n| + |y_{goal} - y_n| \quad (2.3)$$

which is optimal for restrictive grid layouts (Karur, Sharma, et al., 2021). For grid environments that explicitly accommodate uniform diagonal movements along with orthogonal transitions, the Octile distance function computes the search space boundaries as:

$$h_{Octile}(n) = \max(|x_{goal} - x_n|, |y_{goal} - y_n|) + (\sqrt{2} - 1) \cdot \min(|x_{goal} - x_n|, |y_{goal} - y_n|) \quad (2.4)$$

which minimizes structural distance distortion (Karur, Sharma, et al., 2021). Selecting the most structurally appropriate heuristic from these options is critical, as it directly shapes the algorithm's runtime efficiency and exploration footprint (Yao, Lin, et al., 2010).

Modern field robotics applications frequently require expanding these basic distance models into multi-objective frameworks to balance geometric length with environmental safety criteria. This expansion is typically achieved by altering the cost definition of $g(n)$ to explicitly penalize paths that threaten vehicle smoothness, compromise operational safety, or accelerate vehicle energy consumption (Cheng, Liu, et al., 2014; Duchóň, Babinec, et al., 2014; Nazarahari, Khanmirza, et al., 2019). For instance, in maritime

routing domains, the core A* architecture has been improved to seamlessly integrate fluid water currents, traffic separation protocols, and localized maneuverability risks (Zhuang, Zhang, et al., 2019). Similarly, off-road autonomous driving applications utilize modified graph-search models to evaluate instantaneous collision indices and rollover risks across dynamic environmental grid maps (Li, Tang, et al., 2019). However, despite these advancements, the direct, real-time integration of three-dimensional step distances (3DSD) and localized slope magnitudes (SM) as intrinsic components of the cost function for 2.5D agricultural terrain planning, alongside a dedicated adjustable slope weight, remains an unresolved research gap.

To balance exploration breadth and goal-directed guidance during the node search, adaptive cost-weighting schemes are often incorporated into the core evaluation equations. A common linear weighting technique parameterizes the cost equation as:

$$f(n) = g(n) + w \cdot h(n) \quad (2.5)$$

allowing engineers to dynamically tune the trade-off between thorough search exploration ($w < 1$) and focused heuristic guidance ($w > 1$) (Yao, Lin, et al., 2010). Building upon this weighting concept, the framework developed in this thesis introduces an adjustable slope weight parameter (λ) designed to dynamically modulate the relative influence of topographic steepness within a unified multi-terrain cost structure. This approach contrasts sharply with kinematically constrained alternatives like the Hybrid A* algorithm, which couples discrete graph search with continuous motion primitives to guarantee drivable trajectories in constrained spaces (Meng, Yang, et al., 2023). While Hybrid A* effectively secures kinematic feasibility, its standard execution remains restricted to flat 2D occupancy grids, highlighting a lack of explicit, multi-dimensional terrain-cost integration.

2.4.2 Other Path Planning Approaches

While A* and its heuristic variations remain the algorithmic focal point of this research, alternative motion planning paradigms provide essential comparative context regarding multi-dimensional optimization. Sampling-based path planning algorithms, most notably Rapidly-exploring Random Trees (RRT) and Probabilistic Roadmaps (PRM), are highly effective when navigating high-dimensional or geometrically dense configuration spaces, offering mathematical guarantees of probabilistic completeness (Li, Wei, et al., 2020; Sánchez-Ibáñez, Pérez-del Pulgar, et al., 2021). Mechanistically, the standard RRT algorithm explores the vehicle’s workspace by iteratively sampling random spatial coordinates and expanding a tree network from the nearest existing vertex toward these target samples (Li, Wei, et al., 2020; Liu, Wang, et al., 2023). Concurrently, the PRM architecture constructs a continuous topological roadmap by validating interconnected networks

of randomly distributed free-space samples and their visible local neighbors (Sánchez-Ibáñez, Pérez-del Pulgar, et al., 2021). Although these sampling-based approaches effectively mitigate the “curse of dimensionality” that penalizes standard grid-allocation techniques, they introduce significant runtime memory overhead and require exhaustive iteration phases to converge toward true global optimality (Sánchez-Ibáñez, Pérez-del Pulgar, et al., 2021). Consequently, their inherent stochasticity often yields structurally jagged trajectories that necessitate extensive post-processing smoothing to achieve physical execution.

Nature-inspired meta-heuristic algorithms represent an alternative class of global optimization techniques capable of approximating complex trajectories through stochastic search behaviors. Genetic Algorithms (GA), for example, iteratively evolve populations of candidate path solutions through selection, crossover, and mutation mechanisms, providing robust global search capabilities across non-linear spaces (Zhang, Lin, et al., 2018; Zhao, Zheng, et al., 2018). These evolutionary mechanics can be combined with A* cost models to seed populations with high-quality heuristic profiles and accelerate structural convergence (Li, Dong, et al., 2020). Similarly, Particle Swarm Optimization (PSO) maps candidate solutions as discrete particles moving through a multi-dimensional search space, where individual trajectories are guided by a weighted combination of personal historical optima and global swarm best-found states (Zafar & Mohanta, 2018). In agricultural routing, PSO frameworks have been implemented to generate real-time parametric paths that evaluate vehicle kinematic configurations alongside static terrain hazards (Xu, You, et al., 2022). Furthermore, Ant Colony Optimization (ACO) mimics swarm intelligence by deploying agents that collectively identify optimal routes via simulated pheromone deposition (Luo, Wang, et al., 2020; Saeed, Omri, et al., 2022). Although ACO architectures require substantial computational iteration to reach steady-state convergence, optimized variants can dramatically improve processing throughput (Saeed, Omri, et al., 2022). Despite their structural flexibility, the high computational overhead of meta-heuristics often restricts their applicability on resource-constrained embedded hardware.

To bridge global routing with real-time obstacle avoidance, local reactive windowing models are frequently fused with primary graph-search frameworks. The Dynamic Window Approach (DWA) is a popular local path planning algorithm that generates safe velocity commands for mobile robots in real-time by explicitly incorporating the platform’s immediate mechanical and structural dynamics. The algorithm models the search space by sampling admissible acceleration profiles and evaluating short-horizon trajectories against a multi-variant objective function that scores target headings, obstacle clearance distances, and localized velocities (Li & Hu, 2020; Liao, Chen, et al., 2023; Liu, Wang, et al., 2023). By selecting the velocity command that maximizes this reactive function, DWA ensures that global paths adapt dynamically to localized changes without violating kinematic operating limits.

Alternatively, Artificial Potential Field (APF) methodologies model real-time navigation by treating the autonomous platform as a charged particle moving through a continuous virtual force field. Within this mathematical paradigm, the target destination generates an attractive potential gradient that pulls the vehicle forward, while local obstacles project a repulsive potential field that drives the vehicle away from structural hazards (Liu, Wang, et al., 2023; Zafar & Mohanta, 2018). While APF formulations are computationally inexpensive and highly optimized for instantaneous edge processing, they are systematically vulnerable to local minima traps where opposing forces cancel out, causing vehicle oscillations or complete path freeze-ups in cluttered environments (Liu, Wang, et al., 2023; Zafar & Mohanta, 2018). To mitigate these local minima vulnerabilities, hybrid architectures frequently couple APF forces with A* node selection loops to guide the vehicle out of dead-end configurations (Perez-del Pulgar, Perez, et al., 2023). These reactive and force-based local planners provide the essential low-level velocity tracking necessary to physically execute optimized global paths.

2.5 Trajectory Smoothing and Refinement

2.5.1 Need for Smoothing

Global trajectories computed by discrete, grid-based search frameworks like the A* algorithm are fundamentally limited by their structural graph topology, which outputs a sequence of sharp, discontinuous waypoints that result in jagged or angular paths (Karur, Sharma, et al., 2021). Executing these raw piece-wise linear path segments directly is highly undesirable for autonomous ground vehicles because they necessitate abrupt changes in the vehicle's heading vector. These rapid steering adjustments induce severe velocity tracking errors, high mechanical jerk, accelerated fatigue or wear across localized chassis components, elevated electrical power dissipation, and systemic tracking instabilities within low-level feedback controllers (Karur, Sharma, et al., 2021; Xu, 2024; Zhang & Li, 2021). Therefore, implementing an algorithmic post-processing refinement step to transform these piece-wise discrete nodes into continuous, curvilinear trajectories represents a vital prerequisite for elevating vehicle drivability, minimizing tracking errors, and safeguarding platform structural integrity.

2.5.2 B-splines for Path Smoothing

Among continuous parametric optimization formulations, B-spline curves have become a widely adopted method for trajectory smoothing due to their local support prop-

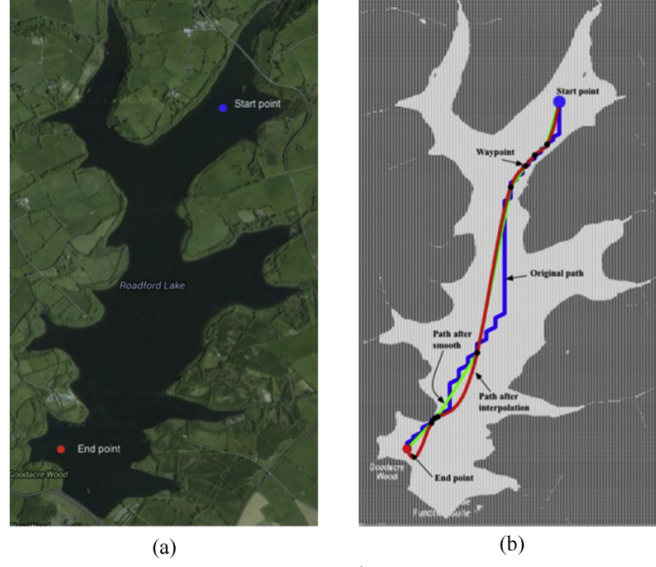


Fig. 11. Off-line path planning simulation result. (a) The satellite map (2.13×3.93 km) of Roadford Lake with marked start and end points. (b) The binary grid map (100×350 grid cells, 1 grid cell column length = 21.3 m and row length = 11.2 m) with trajectories generated using different PSs of the smoothed A* algorithm, i.e. original trajectory without path smoothing, namely the conventional A* algorithm result (blue line), trajectory after applying the LOPS and WRPS, and smoothed trajectory after using the IPS (red line).

Figure 2.4: Illustration of path smoothing applied to a discrete grid-based trajectory.

erty and their ability to generate continuous, differentiable paths (Zhang & Li, 2021). This mathematical formulation allows engineers to alter local curve shapes by manipulating discrete control point matrices without causing global geometric deformations, while simultaneously guaranteeing high-order parametric continuity across the trajectory profile (Nazarahari, Khanmirza, et al., 2019). Specifically, enforcing C^2 continuity—which mathematically secures continuous first and second derivatives—ensures that the vehicle’s bounded lateral and linear accelerations remain smooth and uninterrupted during transit. To bound these parametric properties within defined global terminal states, utilizing open-uniform B-splines is advantageous because their clamped knot vectors force the continuous curve to pass exactly through the initial start and final goal nodes. Translating the discrete waypoints of the graph search into a structured control point matrix produces a kinematically feasible path that prevents sudden actuator saturation or physical freeze-ups during autonomous mobile robot execution (Zhang & Li, 2021).

2.5.3 Validation of Constraints during Smoothing

Although geometric smoothing techniques enhance path aesthetics and velocity tracking performance, a critical research gap exists in rigorously coupling these continuous parametric refinements with concurrent validation of terrain-climbability thresholds and boundary constraints. Simply executing a curvature minimization routine can inadvertently alter the spatial coordinates of a path segment, shifting it onto hazardous topographies that violate the vehicle’s hard physical limits, such as its Maximum Climable

Slope (MCS) (Frey, Hoeller, et al., 2022). Trajectories optimized without these physical considerations may appear visually fluid in the horizontal workspace but remain unsafe or impossible to traverse in non-planar settings (Frey, Hoeller, et al., 2022). Traditional path optimization frameworks typically restrict their safety evaluation loops to basic planar obstacle clearance and curvature smoothness criteria (Karur, Sharma, et al., 2021; Overbye, 2022). However, robust off-road navigation architectures require the integration of continuous rollover hazard evaluations during path selection to guarantee platform stability over severe inclines (Li, Tang, et al., 2019).

This thesis addresses these combined limitations by developing an integrated validation module that directly couples open-uniform B-spline refinement with a high-frequency discrete sampling check. This verification pipeline evaluates the continuous trajectory immediately following or during the curve optimization phase to ensure that the candidate path rigorously adheres to three key boundary parameters:

- **Boundary Compliance:** Keeping the path within defined operational limits.
- **Vegetation Exposure:** Adhering to the limits of the vegetation penalty mask.
- **Maximum Slope Thresholds:** Ensuring no segment exceeds the vehicle’s Maximum Climable Slope (MCS) or other traversability limits.

Enforcing this multi-layered validation during post-processing is crucial for generating truly safe and traversable paths for autonomous off-road agricultural vehicles. Furthermore, data simplification techniques—such as the Douglas-Peucker algorithm or related collinear waypoint suppression methods—are frequently deployed prior to curve fitting to prune redundant nodes. This data reduction minimizes the computational load of the B-spline solver while preserving the underlying topological features of the path (Chen, Lin, et al., 2023; Liao, Chen, et al., 2023).

2.6 Performance Metrics and Evaluation

The comprehensive evaluation of path planning frameworks for autonomous off-road vehicles requires a multi-faceted approach, employing both standard and terrain-specific performance indicators to assess algorithmic utility. Relying solely on a single operational parameter provides an incomplete assessment of a navigation system’s real-world viability. Consequently, modern field robotics literature establishes a verification paradigm that simultaneously benchmarks geometric accuracy, processing velocity, and kinematic feasibility. This multi-layered analytical lens is mandatory for quantifying the exact trade-offs introduced when augmenting a classical routing routine with multi-dimensional terrain awareness.

2.6.1 Standard Performance Indicators

The evaluation of global trajectory profiles begins with analyzing standard spatial and temporal parameters that govern the physical execution of a computed route. The most fundamental indicator is total path length (L), which reflects the geometric distance traversed by the vehicle from its starting configuration to the terminal goal state (Jiang, Xu, et al., 2021; Nazarahari, Khanmirza, et al., 2019; Zhuang, Zhang, et al., 2019). While conventional planar routines calculate this length over a flat projection, terrain-aware models must compute the true three-dimensional (3D) path length (L_{3D}) to account for vertical elevation variations along non-planar surfaces. Concurrently, algorithmic execution runtime and computational efficiency monitor the temporal processing load, logging the total time required to complete graph exploration, cost updates, and validation loops (Duchón, Babinec, et al., 2014; Injarapu & Gawande, 2023; Liu, Meng, et al., 2023; Mac, Copot, et al., 2016; Nazarahari, Khanmirza, et al., 2019). Securing minimized runtime latencies is paramount for ensuring edge-computing feasibility and enabling rapid trajectory adaptation to structural environmental changes. Finally, path smoothness metrics quantify the geometric curvature profiles and derivative continuity of the trajectory, directly influencing tracking stability, low-level actuator wear, vehicle ride comfort, and localized mechanical stress (Karur, Sharma, et al., 2021,?; Li, Tang, et al., 2019; Nazarahari, Khanmirza, et al., 2019). Integrating dedicated steering penalties or post-processing B-spline refinement modules serves to systematically elevate this smoothness index, transforming jagged, node-based waypoints into kinematically executable curvilinear functions (Karur, Sharma, et al., 2021; Zhang & Li, 2021).

2.6.2 Terrain-Specific Metrics (Proxies)

Evaluating autonomous vehicle navigation across unstructured off-road and agricultural domains requires specialized performance metrics capable of quantifying the intricate physical interactions between a mobile platform and a non-planar terrain profile. Relying exclusively on classical geometric indices is insufficient for characterising cross-country mobility, as uniform horizontal displacements can exhibit wildly varying mechanical and temporal costs depending on the local slope gradient or soil configuration. Consequently, modern field robotics research relies on specialized mathematical proxies and physical threshold indices to evaluate a path's true operational viability before physical trajectory tracking.

The temporal efficiency of a non-planar route is evaluated using speed-adjusted travel-time proxies (TTP), which capture the localized velocity reductions that occur due to adverse terrain geometry and topography. While baseline navigation approaches fre-

quently treat vehicle speed as a static, unvarying parameter (Cheng, Liu, et al., 2014), advanced field robotics frameworks incorporate dynamic speed-reduction functions that scale velocity down based on severe slope inclinations, surface roughness, or localized obstacle proximity (Wang, Wang, et al., 2015). Concurrently, optimizing mechanical energy expenditure is a core requirement for ensuring long-endurance operations and economic viability in precision farming (Hameed, Bochtis, et al., 2021). To model these thermal and fuel constraints without the complexity of full multi-body dynamics, architectures implement uphill-selective energy consumption proxies (ECP) within their evaluation loops (Duchón, Babinec, et al., 2014). These energy proxies operate by applying non-linear penalties to positive elevation changes along the path vectors, explicitly reflecting the heightened engine load and power draw required to overcome gravitational resistance during uphill climbs. Finally, an algorithm’s fundamental safety and reliability are quantified via localized success rates and collision avoidance indices, which measure the statistical percentage of completed trials wherein the vehicle reaches its terminal goal state without encountering structural collisions or cross-country immobilization hazards (Lee, Kwon, et al., 2023; Liu, Meng, et al., 2023; Nazarahari, Khanmirza, et al., 2019). In non-planar environments, this validation parameter focuses on assessing whether the trajectory remains bounded within localized regions that comply with the platform’s hard physical Maximum Climable Slope (MCS) limitations.

2.6.3 Simulation Environments

The utilization of virtual simulation environments represents a standard, highly effective methodological convention for prototyping, developing, and evaluating autonomous navigation architectures, particularly when confronting complex and hazardous off-road scenarios (Broggi, Grisleri, et al., 2008; Chilian, Hirschmüller, et al., 2011; Liu, Meng, et al., 2023). High-fidelity virtual simulations facilitate completely reproducible testing protocols across a diverse spectrum of heterogeneous terrain profiles—specifically low-gradient plains, medium-relief hills, and high-relief escarpments—under tightly controlled parameter distributions. This digital sandbox approach bypasses the severe logistical constraints, high capital costs, and physical safety hazards associated with real-world mobile platform deployments (Stentz, Kelly, et al., 2003). Open-source robotics middleware and simulators, most notably Gazebo, are widely used across the field robotics community to safely collect multi-sensor data collections, stress-test prediction networks, and validate global path planning behaviors prior to physical deployment (Liu, Meng, et al., 2023). Consequently, virtual benchmarking serves as a necessary and mathematically sound bridge between pure theoretical algorithmic modeling and physical field testing.

2.7 Summary of Literature and Research Gap

The contemporary body of literature establishes a robust technical foundation in traditional 2D and 3D path planning paradigms, with the classical A* graph-search algorithm serving as a structurally flexible and highly popular choice across diverse navigation sub-problems. The mathematical concept of 2.5D elevation maps is well-established as an effective structural compromise designed to balance rich geometric fidelity with minimized processing latencies in workspaces characterized by prominent vertical elevation changes (Bettencourt, Caramalho, et al., 2025; Xie, Fang, et al., 2020). Furthermore, continuous parametric post-processing formulations, such as B-spline curves, are widely recognized for their mathematical capacity to transform piece-wise linear waypoints into smooth, differentiable, and kinematically suitable trajectories (Zhang & Li, 2021). To quantify the efficacy of these integrated navigation systems, standard validation loops deploy a multi-faceted benchmarking suite that balances structural path lengths and execution runtime efficiency with qualitative criteria tracking curvature smoothness and vehicle collision safety (Karur, Sharma, et al., 2021,?; Mac, Copot, et al., 2016).

However, despite these foundational advancements, several critical technical gaps persist within contemporary literature, particularly concerning the terrain-aware navigation frameworks required for dynamic, complex agricultural off-road environments:

- **Limited Integration of Comprehensive 3D Terrain Features in A* Cost Functions:** While specialized A* variants successfully incorporate isolated environmental modifiers—such as fluid water currents for maritime vessel routing (Zhuang, Zhang, et al., 2019) or general collision and rollover boundary risks for off-road driving (Li, Tang, et al., 2019)—the literature exhibits a clear absence of a unified, explicit cost function operating within a 2.5D A* configuration that simultaneously and directly evaluates physical three-dimensional step distances (3DSD) and localized slope magnitudes (SM). Crucially, the mathematical behavior and proactive path-hazard avoidance generated by combining these 2.5D terrain features with an adjustable slope weight parameter (λ) remains an unaddressed research contribution within the agricultural robotics domain.
- **Insufficient Validation of Constraints during Trajectory Smoothing:** Although continuous geometric refinement techniques like B-splines are widely applied to alleviate the sharp turning maneuvers output by grid-based planners, the consistent, rigorous coupling of these curve-fitting routines with concurrent terrain-climbability threshold enforcement and boundary constraint validation is not extensively addressed. This architectural gap means that smoothed paths that appear visually optimized may actually violate the platform's hard Maximum Climbable Slope (MCS)

or stray outside defined workspace boundaries, leading to unsafe or impossible execution profiles.

- **Lack of Comprehensive Comparative Evaluations Against 2D Baselines:** While the implementation of diverse performance metrics remains standard convention for individual algorithm validation, the literature lacks a comprehensive, direct side-by-side comparative study contrasting a 2.5D terrain-aware A* implementation against a conventional flat 2D A* baseline across a spectrum of heterogeneous synthetic terrain profiles and adjustable slope weight (λ) configurations. This gap in empirical evaluation limits the quantified, mathematically grounded understanding of the specific operational advantages introduced by incorporating multi-dimensional terrain awareness into the core cost function.

This Master's thesis directly responds to these identified scholarly and engineering gaps through an integrated three-part research methodology. By formulating a 2.5D A* algorithm built upon a novel, unified multi-terrain cost function that simultaneously evaluates physical 3DSD, SM, and an adjustable λ , implementing a coupled B-spline smoothing and feasibility validation module, and conducting a comprehensive simulation-based comparative benchmark against a flat 2D baseline, this research delivers targeted, replicable solutions to each identified gap. The next chapter presents the detailed methodology developed to achieve these research objectives.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents a comprehensive and mathematically rigorous exposition of the engineering methodology developed to realize the multi-objective autonomous navigation architecture established in Chapter 1. The algorithmic and validation frameworks designed herein are systematically structured around three core pipeline milestones: first, the mathematical formulation of a two-and-a-half-dimensional (2.5D) A* graph-search algorithm that directly embeds physical three-dimensional step distances (3DSD), localized slope magnitudes (SM), and an adjustable slope weight parameter (λ) into a unified multi-terrain cost function; second, the development of a post-processing B-spline trajectory refinement module that couples continuous curve optimization with a high-frequency discrete sampling check to actively enforce vehicle climbability thresholds and boundary constraints; and third, the implementation of a simulation-based benchmarking suite to evaluate the terrain-aware pathfinder directly against a conventional flat 2D baseline planner. Consequently, this introductory framework serves as the structural foundation for the algorithmic mechanics, kinematic verifications, and empirical data trends detailed throughout this study.

To ensure strict reproducibility and technical transparency within a simulated field environment, this chapter details the explicit computational components, procedural pipelines, and analytical strategies used to validate the system. It describes the synthesized digital assets, the computational processing infrastructure, the step-by-step terrain grid synthesis, the parameter configuration matrices, and the data acquisition protocols. Each architectural choice is systematically justified relative to the specific technical gaps identified in contemporary field robotics literature, ensuring that the resulting metrics directly address the study's core research questions. By executing these sequential validation protocols within a highly controlled virtual sandbox, this methodology establishes a reliable, repeatable, and hardware-scalable environment for quantifying the exact safety and efficiency gains achieved by the proposed terrain-aware framework.

3.2 Research Design and Experimental Architecture

The validation of the proposed motion planning architecture relies on a deterministic, simulation-based experimental design configured around a comparative performance benchmarking paradigm. This systemic architecture is engineered to quantitatively evaluate the operational efficacy of the proposed terrain-aware 2.5D A* algorithm against a conventional flat 2D planar baseline under rigidly controlled and perfectly reproducible environmental conditions. Developing this framework within an entirely virtual software sandbox enables the systematic manipulation of high-dimensional input variables and the precise observation of resulting path characteristics. Consequently, this digital design choice eliminates the unpredictable logistical overhead, physical hardware constraints, and safety hazards inherent in physical off-road agricultural field testing, while guaranteeing absolute mathematical consistency across all evaluation trials.

To thoroughly examine the runtime behavior and optimization limits of the pathfinder, the experimental variables are structured into a multi-tiered evaluation matrix:

- **Terrain Relief Severity Configurations:** The workspace environment is parameterized into three distinct synthetic, deterministic 3D agricultural terrain maps characterizing low-gradient plains, medium-relief hills, and high-relief escarpments. This geometric variation ensures a comprehensive evaluation of the algorithm's topological robustness and structural adaptability across a realistic spectrum of topographical hazards common to unmanaged agricultural fields.
- **Slope Cost Weight Parameter Sweeps (λ):** The operational influence of surface steepness during graph node expansion is analyzed by systematically sweeping the adjustable slope weight parameter (λ) across five discrete values: $\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$. This parameter distribution enables a precise quantitative assessment of how adjusting the penalty weight of localized inclines shifts the trajectory's geometric path optimality.
- **Start-Goal Coordinate Pairs:** To ensure statistical generalizability and eliminate spatial selection bias, three unique, representative start-to-goal coordinate vectors are defined for each distinct terrain profile map.

To maintain internal validity and isolate the performance variations caused by these primary variables, a rigid suite of constant system parameters is uniformly enforced across all computational trials:

- **Grid Cell Size Map Resolution:** All structural 2.5D elevation matrices are strictly maintained at a uniform coordinate resolution of 500×500 cells.

- **Maximum Climable Slope (MCS) Threshold Limit:** A universal hard safety constraint of $\theta_{MCS} = 20^\circ$ (representing an operational vehicle gradient limit of approximately 36% or 0.36 rad) is enforced as a non-negotiable traversability limit. This value models the steepness threshold beyond which the target agricultural platform becomes susceptible to structural slipping or dynamic rollover, rendering such grid coordinates completely impassable.
- **Vegetation Traversal Penalty (VTP) Value:** A fixed scalar cost penalty is uniformly applied to grid coordinates containing dense canopy distributions, reflecting increased locomotion resistance and crop protection boundaries without making the cells strictly forbidden.

For robust performance verification, a total of 15 automated evaluation runs are executed per terrain profile map. This execution matrix comprises evaluating each of the three unique start-goal coordinate vectors across the five distinct slope cost weight settings. This systematic experimental architecture provides the comprehensive empirical datasets necessary to satisfy Objective 3, establishing a clear, multi-variant foundation for performance verification.

3.3 Spatial Data Representations and Environmental Modeling

The environmental modeling pipeline within this computational architecture relies entirely on structured synthetic data arrays that represent the topographical and semantic layers of the agricultural workspace. Because this research is conducted within a virtual development sandbox, these digitized spatial matrices serve as the primary inputs for the navigation stack. Utilizing pure software-driven environmental configurations guarantees that all input geometries remain perfectly controlled, mathematically consistent, and reproducible across all simulation trials.

To construct a high-fidelity workspace for algorithmic evaluation, the environmental datasets are organized into four interconnected core representations:

- **Simulation-Derived 3D Point Cloud Data (.pcd Format):** The foundational spatial data originates as dense three-dimensional (3D) point clouds stored in the standard .pcd file format, mimicking the raw spatial coordinate vectors output by high-resolution field Light Detection and Ranging (LiDAR) sensor suites. These unorganized geometric matrices provide the high-fidelity structural foundation needed to synthesize varied, repeatable terrain profiles without introducing sensor noise or physical environmental variables.

- **Static 2.5D Elevation Grids:** The raw volumetric .pcd datasets are systematically parsed and converted into structured, static two-and-a-half-dimensional (2.5D) elevation grids. This mapping paradigm correlates horizontal planar coordinates (x, y) with a single, discrete altitude value z for each individual grid cell, striking a highly practical structural balance between the spatial blindness of flat 2D models and the massive computational overhead of full 3D volumetric voxel structures. This heightfield representation remains computationally tractable for real-time edge processing while directly supporting the calculation of localized slope gradients and physical three-dimensional step distances.
- **Single-Layer Vegetation Penalty Mask:** Embedded directly as a semantic cost channel within the baseline 2.5D elevation matrices, this localized matrix flags coordinate regions containing dense crop canopies or wild vegetation. These designated cells are uniformly injected with a fixed cost multiplier ($V_{penalty}$), which alters the node expansion priority during graph search by discouraging vehicle translation through dense vegetation zones to protect crop yield and minimize wheel slip.
- **Synthetic Deterministic 3D Agricultural Terrain Profiles:** The operational workspace is explicitly rendered into three distinct digital terrain models classified by relief severity: low-gradient plains, medium-relief rolling hills, and high-relief escarpments. These synthetic agricultural profiles allow for targeted, isolated stress-testing of the pathfinder across a systematically varied spectrum of topographical complexities, reflecting the realistic geometric hazards encountered by heavy machinery in unmanaged fields.

3.4 Computational Infrastructure and Software Ecosystem

The computational validation environment configured for this research comprises a unified software development stack deployed on dedicated processing hardware. Because this methodology bypasses physical mechanical instrumentation, the selected software assets and computing hardware serve as the core infrastructure necessary to compile the navigation logic, execute high-frequency constraint checks, and render dense geometric terrain environments. Standardizing this digital pipeline ensures absolute algorithmic repeatability and eliminates external hardware-induced latency fluctuations across all benchmarking trials.

To manage the processing demands of the multi-objective trajectory simulation, the computational architecture is partitioned into three primary operational tiers:

- **Hardware Processing Infrastructure:** Computational experiments are executed on a dedicated local processing station. This hardware environment provides the

multi-threaded throughput and memory bandwidth required to run dense graph-search operations, spatial heightfield conversions, and continuous B-spline matrix solvers without data throttling.

- **Core Software Environment and Specialized Libraries:** The software architecture is developed within a native Linux/Ubuntu ecosystem using Python as the primary programming language for all algorithmic implementations, spatial simulations, and terminal data parsing. Python is chosen due to its extensive scientific computing ecosystem and robust interpreter performance during rapid prototyping cycles. Within this environment, localized array manipulations and mathematical operations are accelerated using several domain-specific libraries:
 - **NumPy:** Deployed to execute high-performance vectorized cell matrix manipulations, coordinate translations, and fundamental Euclidean distance equations.
 - **SciPy:** Utilized for advanced mathematical computing pipelines, specifically managing multi-dimensional spatial filtering during terrain synthesis and solving B-spline knot vectors during path refinement.
 - **Matplotlib:** Integrated to visualize data structures by rendering interactive 2.5D heightmaps, comparative trajectory curves, and analytical performance histograms.
- **Downstream Framework Compatibility:** All mapping outputs, cost-map transformations, and pathfinding logic are structurally engineered to adhere to data types compatible with the Robot Operating System 2 (ROS2) and Autoware architectures. This design choice ensures that the developed software modules can transition into production-level autonomy stacks for hardware-in-the-loop (HIL) testing or integration into complex autonomous vehicle ecosystems.

3.4.1 Reference Physical Platform

To ground the virtual workspace boundaries within realistic operational limits, the kinematic specifications of the Pix Kit autonomous field platform are integrated as design constraints during software development. While no physical vehicle trials are conducted within the scope of this thesis, the platform’s real-world parameters—specifically its wheelbase geometry, track width, center of mass elevation, and electrical powertrain limits—are used to establish the hard Maximum Climable Slope (θ_{MCS}) limits enforced within the cost function. Aligning the simulation parameters with a commercial-grade autonomous chassis ensures that the generated global trajectories remain safe, practical, and highly relevant for future physical agricultural deployments.

3.5 Algorithmic Pipeline and Implementation Procedures

3.5.1 Point Cloud Rasterization and 2.5D Grid Generation

The initial phase of the methodology establishes the foundational environmental workspace by transforming unstructured, high-density three-dimensional (3D) spatial coordinate vectors into a discretized topology optimized for graph-search operations. This data transformation process operates through three tightly coupled execution steps:

- **Volumetric Data Input:** The pipeline initializes by loading synthetic 3D point cloud datasets (.pcd format) characterizing the three benchmark agricultural terrain profiles (low-gradient plains, medium-relief rolling hills, and high-relief escarpments). These unorganized point matrices provide the true baseline geometric information of the simulated agricultural workspace.
- **Planar Projection and Rasterization:** The raw spatial points are orthogonally projected onto a horizontal coordinate ground plane parameterized by horizontal coordinates (x, y) and discretized into a uniform, static two-and-a-half-dimensional (2.5D) grid array with a fixed dimension resolution of 500×500 cells. For each individual grid cell boundary, a single, discrete altitude value z is calculated by extracting and mapping the maximum elevation value from the subset of raw point clouds bounded within that cell's physical spatial footprint. This conservative mapping choice acts as a safety-first bounding envelope, ensuring that micro-topographical height variations, surface ridges, and localized obstacle tops are fully captured to mitigate clearance hazards while completely removing the excessive computational overhead of full volumetric voxel grids.
- **Semantic Costume Mask Integration:** Following heightfield generation, a specialized single-layer semantic vegetation penalty mask is structurally superimposed onto the baseline 2.5D elevation grid. This semantic overlay cross-references coordinate indices to isolate cells containing dense crop canopies or unmanaged ground cover, uniformly injecting a fixed scalar cost modifier ($V_{penalty}$) into the cell's properties. By embedding this soft cost boundary directly into the terrain data array, the local pathfinder is mathematically forced to treat dense vegetation as a high-resistance navigation zone during runtime node expansion, protecting crop yields and minimizing platform slip.

3.5.2 Formulation of Terrain-Aware Step Traversal Cost (STC)

To achieve a highly responsive, topology-aware graph expansion under the modified A* path planning architecture, a novel terrain-aware Step Traversal Cost (STC) function is formulated to evaluate edge weights between adjacent nodes. This multi-variant cost model directly satisfies the requirements of Objective 1 by integrating physical three-dimensional step distances (d_{3D}), localized slope magnitudes (θ), a semantic vegetation modifier ($V_{penalty}$), and an adjustable slope weight parameter (λ) into a unified optimization metric. For a discrete translation step from a current parent cell P to an adjacent candidate cell C within an 8-connected neighborhood, the instantaneous traversal cost is modeled via the following structural equation:

$$STC(P, C) = d_{3D}(P, C) + \lambda \cdot \theta(P, C) + V_{penalty}(C) \quad (3.1)$$

where the constituent geometric and semantic sub-functions are defined and mathematically bounded as follows:

- **Three-Dimensional Step Distance (d_{3D}):** This parameter quantifies the true 3D Euclidean distance separating the volumetric geometric centers of parent cell P and candidate cell C . Unlike conventional planar planners that rely on 2D horizontal projections—which treat translations over sloped and flat planes identically—the d_{3D} function incorporates heightfield differentials to capture the real physical travel distance and mechanical effort:

$$d_{3D}(P, C) = \sqrt{(x_C - x_P)^2 + (y_C - y_P)^2 + (z_C - z_P)^2} \quad (3.2)$$

- **Local Slope Magnitude (θ):** This metric represents the topographic steepness or spatial gradient established between the center coordinates of cells P and C . The slope magnitude is calculated as the arctangent of the ratio of the vertical elevation differential (Δz) to the horizontal distance (d_{2D}) separating the cell centers:

$$\theta(P, C) = \arctan \left(\frac{|z_C - z_P|}{d_{2D}(P, C)} \right) \quad (3.3)$$

Elevated θ profiles indicate high-relief terrain features that introduce immediate kinematic tracking hazards, track slippage, or platform instability.

- **Adjustable Slope Weight Parameter (λ):** This scalar coefficient scales the mathematical impact of the local slope magnitude on the total accumulative step cost. Swept systematically across five discrete settings within this research design: $\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$. A higher λ value heavily penalizes steep inclines, forcing the graph-search routine to bypass prominent elevation changes in favor of lat-

eral contour paths. This tunability allows field engineers to dynamically realign the navigation system based on vehicle power limits, soil conditions, or operational preferences (e.g., minimizing fuel consumption versus minimizing overall transit distance).

- **Vegetation Traversal Penalty ($V_{penalty}$):** This binary semantic filter evaluates whether candidate cell C falls within the boundaries of the single-layer vegetation mask. If a vegetation hazard is detected, a fixed scalar penalty is added directly to the edge calculation; otherwise, the term is nullified ($V_{penalty} = 0$). This soft constraint influences trajectory selection by guiding the vehicle away from unmanaged crops or dense brush without designating those zones as strictly impassable obstacles.

3.5.3 Implementation of Modified 2.5D A* Search Algorithm

The core graph-search architecture underpins a modified A* path planning algorithm configured to execute across an 8-connected grid network topology. This implementation operates directly on the compiled elevation matrices, using the previously formulated Step Traversal Cost (STC) to dynamically alter node cost evaluation. By integrating real-time geometric and semantic terrain layers into the primary search loop, the pathfinder proactively routes the mobile robot around untraversable topographical features. The deterministic graph search evaluates and registers candidate nodes by incorporating these continuous cost metrics into the standard A* heuristic evaluation framework:

$$f(n) = g(n) + h(n) \quad (3.4)$$

where the algorithmic cost components and hard safety filters are integrated as follows:

- **Cost Function Integration ($g(n)$):** The cumulative cost component $g(n)$, which tracks the exact cost accrued from the initial start state to the current node n , is calculated by summing the discrete, terrain-aware step traversal costs (STC) incurred during each successive node transition:

$$g(n) = \sum_{i=0}^k STC(P_i, C_i) \quad (3.5)$$

By mapping $g(n)$ directly to the STC function, the algorithm intrinsically considers the combined effects of the three-dimensional step distance (d_{3D}), localized slope magnitude (θ), the user-defined slope weight parameter (λ), and the semantic vegetation traversal penalty ($V_{penalty}$) during the live graph exploration phase.

- **Heuristic Function ($h(n)$):** The heuristic function $h(n)$ provides a computationally efficient, goal-directed bias by estimating the remaining traversal cost from the current node n to the terminal goal state. To guarantee mathematical admissibility (ensuring the planner never overestimates the true remaining cost) and secure global optimality, $h(n)$ is parameterized as the true three-dimensional (3D) Euclidean distance between node center coordinates:

$$h(n) = \sqrt{(x_{goal} - x_n)^2 + (y_{goal} - y_n)^2 + (z_{goal} - z_n)^2} \quad (3.6)$$

This spatial formulation ensures that vertical height field variations between the active search front and the goal coordinate are accounted for, generating an accurate and admissible heuristic envelope.

- **Hard Maximum Climbable Slope (MCS) Safety Filter:** To prevent the pathfinder from generating trajectories that violate the hard physical operating limits of the agricultural mobile platform, a proactive safety constraint checker is embedded directly within the neighbor expansion loop. Prior to pushing a valid neighbor node onto the Open List, the local slope magnitude (θ) between parent cell P and candidate cell C is evaluated against the non-negotiable vehicle threshold (θ_{MCS}):

$$\text{if } \theta(P, C) > \theta_{MCS} \Rightarrow \text{node } C \text{ is pruned (impassable)} \quad (3.7)$$

Any potential step that exceeds this boundary is immediately classified as un-traversable, forcing the algorithm to treat the target coordinate as a rigid structural obstacle and prune it from the active search space. This proactive filtering guarantees that all generated raw paths respect the structural stability limits of the underlying vehicle chassis before entering the trajectory refinement stage.

3.5.4 Conventional 2D Baseline Path Planning

To establish a rigorous mathematical baseline for the comparative analysis executed in Objective 3, a conventional planar A* algorithm is implemented as a benchmark tracking routine. This control planner operates exclusively on a flattened, two-dimensional matrix projection of the environment, completely stripped of continuous vertical height topologies and localized surface gradients. By executing graph exploration over this simplified configuration layout, the baseline pathfinder generates standard piecewise linear trajectories that reflect traditional coordinate routing methodologies. The planar search routine evaluates grid nodes within a standard 8-connected pixel neighborhood using the baseline evaluation function:

$$f_{2D}(n) = g_{2D}(n) + h_{2D}(n) \quad (3.8)$$

where the internal cost-tracking parameters and obstacle handling mechanisms are formulated as follows:

- **Planar Cost Function (g_{2D}):** The cumulative tracking cost g_{2D} is parameterized strictly by the horizontal 2D Euclidean distance separating adjacent cell centers. This formulation completely omits altitude changes (Δz), treating translations across steep hills and flat plains as energetically and temporally identical:

$$g_{2D}(n) = \sum_{i=0}^k \sqrt{(x_{C_i} - x_{P_i})^2 + (y_{C_i} - y_{P_i})^2} \quad (3.9)$$

- **Planar Heuristic Function (h_{2D}):** The goal-directed guidance heuristic h_{2D} is similarly constrained to the horizontal coordinate plane, computing the absolute straight-line distance from the active node n to the destination coordinates without factoring in target peak elevations:

$$h_{2D}(n) = \sqrt{(x_{goal} - x_n)^2 + (y_{goal} - y_n)^2} \quad (3.10)$$

- **Binary Obstacle Handling:** To protect the mobile platform from immediate collision failures, macro-topographical features and slope hazards are managed using a strict binary segmentation mask. If a grid cell's maximum elevation profile natively exceeds the vehicle's predefined Maximum Climable Slope threshold (θ_{MCS}), that coordinate is classified as a hard barrier and pruned from the search envelope. However, unlike the proposed 2.5D A* algorithm, intermediate slope variations, continuous terrain roughness indices, and semantic vegetation costs do not influence the edge weights of passable cells. The baseline planner treats all non-obstacle terrain as a uniform, zero-cost plane, highlighting the technical limitations that lead to tracking lag and mechanical stress when executing planar paths over non-planar agricultural surfaces.

3.5.5 Open-Uniform B-spline Trajectory Smoothing with Feasibility Checks

Following the generation of the initial, discrete path waypoints from either the proposed 2.5D A* or the baseline 2D A* graph-search algorithms, the trajectory is passed into a specialized post-processing refinement module. This pipeline phase directly satisfies Objective 2 by optimizing path curvature while actively validating safety and environmental constraints across the continuous domain. This dual-purpose refinement process

operates through two sequential operations:

- **Open-Uniform B-spline Curve Fitting:** The ordered set of discrete waypoints output by the pathfinder is treated as a topological control point matrix to generate an open-uniform B-spline curve. B-spline formulations are chosen for trajectory optimization because of their local support property, which allows for localized geometric adjustments without causing global shape deformation. To achieve smooth, continuous vehicle velocities and bounded lateral acceleration profiles during physical trajectory tracking, the refinement module enforces strict C^2 continuity across the parametric curve. Furthermore, using an open-uniform knot vector configuration clamps the terminal endpoints of the spline function, forcing the continuous curve to pass exactly through the original start and goal coordinate coordinates to maintain global boundary alignment.
- **High-Frequency Discrete Sampling Checks:** To ensure that the continuous geometric refinement has not inadvertently shifted trajectory segments onto unsafe terrains, the optimized B-spline curve undergoes a rigorous validation process. The parametric curve equation is evaluated at uniform, high-frequency intervals to generate a dense array of discrete spatial tracking points. At each sampled coordinate along the continuous path, the system executes three synchronized validation routines:
 - **Maximum Slope Threshold Validation:** The instantaneous localized slope derivative of the smoothed trajectory segment is computed and cross-referenced against the vehicle's hard operating limit (θ_{MCS}). If any individual sampled point exhibits a localized incline exceeding this threshold, the entire candidate trajectory is flagged as invalid.
 - **Workspace Boundary Compliance:** The spatial coordinates of each sample point are evaluated against the global configuration space dimensions to verify that the continuous path remains fully contained within the authorized operational tracking boundaries of the agricultural terrain map.
 - **Vegetation Exposure Verification:** The sample coordinates are mapped directly onto the underlying semantic mask layer to track and limit cumulative vehicle exposure to dense vegetation zones, protecting crop rows from tire damage and preventing wheel slippage.

This multi-layered validation routine acts as a strict safety filter during the post-processing phase, preventing the generation of trajectories that are kinematically smooth but physically untraversable or unsafe for field operation.

3.5.6 Parametric Data Logging

To ensure an objective and comprehensive benchmarking process, the optimization framework is coupled with an automated parametric data-logging module. Upon the successful completion and validation of each trajectory generated by both the proposed 2.5D A* terrain-aware algorithm and the conventional 2D A* baseline planner, the logging system intercepts the workspace state and captures a high-fidelity quantitative profile. This data package tracks five primary operational metrics: exact three-dimensional path length (L_{3D}), speed-adjusted travel-time proxies (TTP), computational execution run-times, uphill-selective energy consumption proxies (ECP), and the adjustable slope weight coefficient (λ). These recorded performance vectors are systematically dumped into structured flat files (.csv format), creating a clean empirical ledger that supports direct statistical cross-evaluation during the performance verification stage.

3.6 Digital Map Synthesis and Environmental Configuration

The generation of standardized, controlled, and mathematically reproducible workspaces is a necessary prerequisite for isolating the optimization performance of the underlying pathfinding routines. Because this research bypasses physical field deployments, the environmental mapping configurations are generated synthetically to establish clear topographical benchmarks. This software-driven synthesis process eliminates chaotic real-world variables—such as dynamic weather conditions, localized GPS multipath interference, or soil shear strength fluctuations—while ensuring that all input maps share a uniform cell resolution and data layout to maximize internal validation.

To build out the evaluation workspace, the environmental data synthesis pipeline executes across four tightly integrated configuration phases:

- **High-Fidelity Point Cloud Generation:** The synthesis begins with programmatically generating dense, three-dimensional (3D) point cloud matrices stored in the standard .pcd file format. These point configurations are mathematically generated to build three distinct agricultural terrain profiles classified by relief severity: low-gradient plains, medium-relief rolling hills, and high-relief escarpments. This geometric step locks in the baseline spatial characteristics for each experimental scenario, ensuring that the primary elevation data remains completely controlled and reproducible across independent testing blocks.
- **Coordinate Rasterization and 2.5D Conversion:** Each synthetic 3D point cloud is parsed through a coordinate projection and rasterization script. This routine compresses the unorganized spatial data points into a static, two-and-a-half-dimensional

(2.5D) elevation matrix with a fixed resolution of 500×500 cells. To maintain absolute safety-first consistency, each individual 2.5D grid coordinate is assigned a single elevation value z derived by extracting the maximum elevation value from the underlying point cloud subset bounded within that cell’s spatial footprint, capturing localized surface peaks to mitigate platform clearance hazards.

- **Semantic Vegetation Mask Overlay:** A single-layer semantic penalty mask is programmatically synthesized and overlaid onto the baseline 2.5D heightfields. This matrix matches the global grid dimensions to flag specific coordinates as heavily vegetated zones, such as unmanaged crop rows or dense undergrowth. These designated coordinate blocks are injected with a fixed cost multiplier ($V_{penalty}$), forcing the active search front to treat the vegetation zones as high-resistance paths.
- **Deterministic Start-Goal Vector Definitions:** To prevent spatial selection bias, three unique start-to-goal coordinate vectors are defined for each of the three terrain profile maps. These operational coordinate pairs are chosen to represent varied navigational challenges—such as routing diagonally across steep ridges or skirting extensive vegetation zones—providing a diverse set of test cases. This structured configuration ensures that all digital environmental configurations are uniform in format, controlled in their characteristics, and optimized for automated input into the path planning framework.

3.7 Verification Framework and Benchmarking Protocols

The operational performance and structural reliability of the navigation architecture are evaluated using a deterministic, simulation-based path planning benchmarking suite. Because this research bypasses physical mechanical deployment, the verification loop relies on software-driven performance benchmarking to isolate internal system behaviors from external environmental variables. This rigorous analytical design choice replaces unpredictable laboratory variables with highly reproducible, software-defined testing boundaries, providing clear, uncompromised empirical evidence that directly addresses Objective 3.

To systematically map out the operational boundaries and optimization trade-offs of the pathfinder, the evaluation pipeline is structured into three primary benchmarking domains:

- **High-Fidelity Virtual Environment Testing:** All algorithmic iterations, coordinate expansion loops, constraint checking steps, and structural path visualizations are conducted within a unified virtual simulation sandbox. This fully digitized environment supports precise replication of experiments, isolation of independent de-

sign variables, and targeted edge-case testing without the systemic tracking noise, GPS signal degradation, mechanical wear, or structural safety hazards inherent to physical off-road autonomous mobile robot trials.

- **Direct Comparative Benchmarking:** The core verification routine involves a direct comparative analysis between the proposed terrain-aware 2.5D A* pathfinder and the conventional flat 2D A* planar baseline algorithm. By testing both planners over identical coordinate origins under uniform map criteria, this control architecture establishes a clear mathematical contrast. This layout is necessary to isolate and quantitatively demonstrate the precise geometry, runtime, and energy consumption changes gained through multi-dimensional terrain awareness.
- **Systematic Multi-Variant Experimental Matrix:** To gather exhaustive performance profiles and eliminate data anomalies, the benchmarking routine executes across a pre-calculated, automated test grid:
 - **Topographical Testbeds:** The evaluation spans all three synthetic terrain profiles (low-gradient plains, medium-relief rolling hills, and high-relief escarpments), testing the system across varying levels of vertical relief.
 - **Boundary Pair Vectors:** Within each individual terrain map, three unique, representative start-to-goal coordinate vectors are used to test a diverse set of obstacle layouts and elevation obstacles.
 - **Parametric Parameter Sweeps:** For each distinct start-to-goal vector, the proposed 2.5D A* algorithm is iteratively executed across five discrete slope weight allocations: $\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$.

This systematic distribution yields exactly 15 automated evaluation runs per terrain profile map, totaling 45 unique data points for the proposed pathfinder, which are mirrored by corresponding control trials using the planar 2D A* baseline.

3.7.1 Performance Characterization Suite

For every path generated across the 45 unique test configurations and their corresponding baseline trials, a comprehensive suite of quantitative performance metrics is calculated and recorded. Rather than evaluating path aesthetics qualitatively, this thesis utilizes numerical data logging to evaluate trajectory viability across five primary areas: total three-dimensional path length (L_{3D}), speed-adjusted travel-time proxies (TTP), overall computational execution runtime, uphill-selective energy consumption proxies (ECP), and path smoothness indices derived from continuous B-spline derivative variations. This structured data acquisition strategy provides the foundational datasets required to verify

the efficiency, structural safety, and real-time edge-computing viability of the proposed framework, ensuring total alignment with the final objectives of this study.

3.8 Automated Data Collection Protocols

Data collection is executed as an automated, high-frequency process that hooks directly into the core navigation pipeline during simulation trials. Upon the validation of a trajectory generated by either the proposed terrain-aware 2.5D A* algorithm or the conventional 2D A* baseline, the execution loop pauses to dump the complete internal state vector of the pathfinder. This systematic logging routine generates an exhaustive, multi-variant dataset completely free from human transcription errors or sampling irregularities. By capturing synchronized spatial, temporal, and mechanical variables simultaneously, this collection architecture provides the foundational data needed to run detailed trade-off analyses across varying slope weight (λ) configurations.

To support the comparative evaluations detailed in Objective 3, five distinct response properties are precisely computed, mapped, and logged for each individual trajectory:

- **Total Three-Dimensional Path Length (L_{3D}):** This metric registers the absolute cumulative Euclidean length of the path segments in 3D space. By incorporating vertical coordinate changes along the heightfield, this parameter captures the true physical distance traveled over undulating agricultural landscapes rather than relying on a shortened horizontal projection.
- **Speed-Adjusted Travel-Time Proxies (TTP):** This value maps mission duration by applying a dynamic deceleration function to the vehicle’s baseline velocity vector. The calculation factors in predictable speed reductions caused by adverse surface conditions—specifically penalizing steep localized gradients and high-density vegetation zones—providing a highly realistic estimation of real-world operational timing.
- **Algorithmic Execution Runtime:** This temporal metric logs the total computational processing time required to output a traversable trajectory. It is measured from the initial loading of the start-goal vector to the final termination of the post-processing loop, encompassing both the heuristic A* graph-search phase and the subsequent B-spline refinement and discrete sampling checks. Monitoring this parameter is critical for assessing the architecture’s suitability for low-latency, real-world edge deployment on embedded vehicle computers.

- **Uphill-Selective Energy Consumption Proxy (ECP):** This property estimates overall battery or fuel depletion by tracking the cumulative mechanical work required to navigate the surface profile. The underlying cost function applies a directional penalty to positive elevation changes ($\Delta z > 0$), modeling the heightened torque requirements and electrical power draw needed to overcome gravitational resistance during uphill climbs.
- **Mean Turn Angle Reductions:** This metric provides a geometric index of path smoothness by computing the mathematical average change in orientation angles across successive trajectory waypoints. By measuring the exact difference between the sharp, jagged turn angles of the raw A* output and the smooth curvature of the refined B-spline path, this variable quantifies the effectiveness of the smoothing module in minimizing lateral tire scrubbing and high-frequency steering actuator wear.

3.9 Data Analysis and Statistical Framework

The compiled empirical dataset is subjected to a comparative multi-metric statistical evaluation using a Python-based processing pipeline. Rather than relying on isolated qualitative observations, this analytical framework uses quantitative descriptive and inferential tracking models to evaluate the system. This structured data analysis phase provides the mathematical validation required to address Objective 3, translating raw logged simulation data into verifiable performance trends regarding path safety, efficiency, and real-time edge viability.

To extract meaningful engineering conclusions from the 45 unique simulation configurations and their corresponding baseline runs, the data analysis framework implements four interconnected operations:

- **Direct Comparative Performance Analysis:** Trajectory metrics are grouped by terrain profile severity (low-gradient, medium-relief, and high-relief) and cross-referenced against their exact start-to-goal vector pairs. The performance parameters achieved by the proposed terrain-aware 2.5D A* pathfinder are directly compared against the flat 2D A* planar control baseline using percentage-difference matrices. This localized alignment isolates the precise geometric, temporal, and energetic trade-offs caused by removing vertical topology during global path planning.
- **Parametric Sensitivity Analysis of Slope Cost Weight (λ):** The operational influence of the adjustable slope weight coefficient is systematically mapped by tracking variations in response properties across the discrete sweep spectrum ($\lambda \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$).

This sensitivity analysis monitors the rate of change for L_{3D} , TTP, ECP, and mean turn angle reductions within each terrain class. Characterizing these trends provides a clear look at how path behavior alters when prioritizing slope-avoidance safety over shortest-path length.

- **Descriptive and Inferential Statistical Modeling:** Baseline data arrays are first characterized using descriptive statistics, computing the arithmetic mean ($\bar{x}$) and standard deviation (σ) for each parameter subset to establish central tendency and data dispersion. Where applicable, inferential statistical models are deployed to determine mathematical significance; specifically, two-tailed Student's t-tests are used for direct 2.5D A* versus 2D A* comparisons, while one-way Analysis of Variance (ANOVA) routines evaluate variance across the five distinct λ settings to ensure observed differences are due to algorithmic design rather than random simulation variance.
- **Graphical Visualization and Data Rendering:** Matplotlib is used to construct publication-grade graphical representations that visually communicate identified performance boundaries and structural trends. The data pipeline generates clustered bar charts to contrast multi-metric performance across planners, line graphs to depict parametric sensitivity shifts as a function of changing λ , and two-dimensional spatial projections overlaid with continuous trajectory curves to illustrate path deviation relative to underlying terrain gradients.

3.10 Architectural Reliability, Repeatability, and Error Control

Securing the systemic reliability, structural repeatability, and strict error control of all experimental outcomes is fundamental to establishing the scientific rigor and validity of this research. Because empirical evaluation is executed within a purely digital workspace, error propagation is managed by neutralizing stochastic software behaviors and environmental variables. This highly controlled engineering design choice ensures that any recorded variances in the performance ledger are exclusively a function of algorithmic design and parameter tuning, satisfying the strict documentation requirements of reproducible technical research.

To maintain rigid internal validity and enforce strict error control across the experimental framework, the methodology implements five core control mechanisms:

- **Deterministic Simulation Environment Architecture:** The software stack operates within an entirely deterministic virtual simulation architecture. Because all stochastic search heuristics are omitted from both the proposed 2.5D A* algorithm

and the 2D A* baseline, providing identical inputs—specifically matching the height-field matrices, start-to-goal vector pairs, and scaling coefficients—consistently yields identical path trajectories and processing costs. This algebraic determinism supports robust comparative analysis and guarantees a mathematical repeatability coefficient of 1.0.

- **Synthetic Topographical Benchmarks:** Utilizing synthetic, deterministic 2.5D agricultural terrain profiles (low-gradient, medium-relief, and high-relief) eliminates chaotic environmental variability. Unlike physical field deployments characterized by unpredictable wind vectors, varying soil moisture content, and localized GPS signal multi-path reflections, these static digital elevation grids maintain structural geometry across all evaluation trials. This spatial stability ensures that observed changes in vehicle velocity or power draw proxies are solely attributable to the planner’s routing decisions.
- **Standardized Automated Experimental Matrix:** The verification pipeline relies on a predefined, automated testing script that loops through the evaluation grid. Automating this execution matrix eliminates human sampling bias and prevents human operational errors during data collection, ensuring that the entire evaluation pipeline can be replicated on any compatible computing infrastructure.
- **Rigidly Maintained Global Fixed Parameters:** All baseline operating parameters are locked across independent testing blocks using global software constants. This includes maintaining a uniform coordinate map dimension resolution of 500×500 cells, a non-negotiable threshold for the vehicle’s maximum climbable slope (θ_{MCS}), and a uniform vegetation traversal penalty ($V_{penalty}$). Holding these boundary layers constant removes confounding variables and isolates the precise sensitivity profile of the adjustable slope weight (λ).
- **Control Group Baseline Integration:** The conventional flat 2D A* planar pathfinder serves as an algorithmic control group within this research design. Benchmarking the proposed 2.5D A* terrain-aware algorithm against this flat baseline isolates the specific performance changes gained from multi-dimensional terrain awareness, preventing overestimation of algorithmic utility.

While virtual simulation models represent a necessary geometric abstraction of physical field dynamics, this highly controlled and repeatable testing framework establishes a mathematically sound environment for validating path planning logic. It provides a reliable engineering foundation for verifying theoretical optimization bounds and identifying promising navigation advancements before migrating the software modules onto physical agricultural platforms.

3.11 Summary of Methodology

This chapter has presented a detailed and mathematically rigorous overview of the engineering methodology developed to achieve the research objectives of this Master’s thesis. The underlying experimental design is structured around a deterministic, simulation-based comparative study engineered to evaluate the operational efficacy of the proposed terrain-aware 2.5D A* path planning framework against a conventional flat 2D A* planar control baseline. The environmental architecture relies on high-fidelity synthetic 3D point cloud datasets rasterized into static 2.5D elevation matrices using a conservative maximum elevation mapping envelope. These heightfields are seamlessly integrated with programmatically generated single-layer semantic vegetation penalty masks to construct a controlled testing suite spanning three distinct topographical relief profiles: low-gradient plains, medium-relief rolling hills, and high-relief escarpments. All algorithmic modules, cost-map transformations, and statistical evaluation scripts are implemented natively within a Python environment utilizing the high-performance computing capabilities of the NumPy, SciPy, and Matplotlib libraries, ensuring structural data type compatibility with the Robot Operating System 2 (ROS2) and Autoware autonomy frameworks.

To provide a transparent, step-by-step validation tracking loop, the core computational procedure of the navigation stack is executed through five tightly coupled execution steps:

1. **Spatial Grid Processing:** Compressing unorganized volumetric point cloud vectors (.pcd format) into structured, uniform 500×500 cell coordinate arrays.
2. **Cost Function Formulation:** Establishing a novel, physics-informed Step Traversal Cost (STC) framework that intrinsically combines three-dimensional step distances (d_{3D}), localized slope magnitudes (θ), an adjustable slope weight parameter (λ), and a semantic vegetation traversal penalty ($V_{penalty}$).
3. **Heuristic Global Graph Search:** Deploying a modified 2.5D A* grid-search pathfinder that maps the STC directly to the cumulative cost function $g(n)$, while using an inline Maximum Climable Slope (θ_{MCS}) safety filter to proactively prune hazardous coordinates from the active search space.
4. **Continuous Trajectory Refinement:** Transforming discrete, piece-wise linear waypoints into a C^2 continuous curve using an open-uniform B-spline module coupled with high-frequency discrete sampling checks to prevent geometric smoothing from violating hard physical or environmental thresholds.
5. **Baseline Control Execution:** Routing identical coordinate origins across a flattened, zero-cost representation of the environment using a conventional 2D A* baseline algorithm to isolate performance differences.

The data acquisition system automatically logs five primary response properties for every generated path segment: total three-dimensional path length (L_{3D}), speed-adjusted travel-time proxies (TTP), overall computational execution runtime, uphill-selective energy consumption proxies (ECP), and mean turn angle reductions. This compiled data ledger is subjected to descriptive and inferential statistical evaluations—specifically utilizing percentage-difference matrices, Student’s t-tests, and one-way ANOVA routines—to mathematically isolate the performance advantages of multi-dimensional terrain awareness and chart the system’s sensitivity to variations in the slope weight coefficient (λ). This highly controlled, repeatable methodology directly addresses the technical research gaps identified in Chapter 2, providing a robust, scientifically sound framework to answer all stated research objectives.

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APPENDICES

APPENDIX A

Research publications

Below are lists of contributions in form of publications and presentations:

Book series

- M.R. Othman, J.U. Repke & G. Wozny. 2010. "Incorporating Negative Values in AHP Using Rule-Based Scoring Methodology for Ranking of Sustainable Chemical Process Design Options". Computer Aided Chemical Engineering. Vol. 28, pp. 1045-1050A.

Journal publication

- M.R. Othman, J.U. Repke, G. Wozny & Y. Huang. 2011. "Sustainability Assessment & Decision Making for Sustainable Chemical Process Option". To be appear in Chinese Journal of Chemical Engineering.
- M.R. Othman, J.U. Repke, G. Wozny & Y. Huang. 2010. "A Modular Approach to Sustainability Assessment and Decision Support for Chemical Process Design". Industrial & Engineering Chemistry Research. 49 (17), pp. 7870-7881.

APPENDIX B

Conference presentations

Conference presentation

- M.R. Othman, L. Hady, J.U. Repke & G. Wozny. 2011. "Introducing Decision Making Methodology into Chemical Engineering Education". 8th European Congress of Chemical Engineering. 25-29 September 2011. Berlin, Germany. POSTER
- M.R. Othman, J.U. Repke & G. Wozny. 2011. "Selection of Sustainable Chemical Process Design using ANP: A Biodiesel Case Study". International Symposium on the Analytic Hierarchy Process (ISAHP). 15-18 June 2011. Sorrento, Naples, Italy. ORAL
- M.R. Othman, J.U. Repke & G. Wozny. 2010. "Incorporating Negative Values in AHP Using Rule-Based Scoring Methodology for Ranking of Sustainable Chemical Process Design Options". 20th European Symposium on Computer Aided Process Engineering (ESCAPE 20). 6-9 June 2010. Ischia, Naples, Italy. ORAL
- M.R. Othman, J.U. Repke, G. Wozny & Y. Huang. 2010. "Sustainability Assessment and Decision Making for Sustainable Chemical Process Option". 2nd International Symposium on Sustainable Chemical Product & Process Engineering (ISS-CPPE2). 9-12 May 2010. Hangzhou, China. ORAL & POSTER
- M.R. Othman, J.U. Repke, G. Wozny & Y. Huang. 2009. "A Modular Approach to Sustainability Assessment & Decision Support in Chemical Process Design". AIChE: 1st International Congress on Sustainability Science & Engineering (ICOSSE). 10-13 August 2009. Cincinnati, Ohio, USA. ORAL